The Big Pool Bangs Theory

Getting Rid of Dark Matter and Comic Inflation

Abstract. The universe is a set of concentric rings of identical Planck-mass relics obeying different mechanics in each ring; we live in the innermost, Kernel 0,0,0, the only ring with light: the Big Apple. Dark matter needs no new particle: it is the Planck-mass relics that survive the grinding in Kernel 0,0,0 — the trace fraction $f = T_{\text{eq}}/T_{\text{conv}}$ left when relic mergers, which convert relic mass to light, stop keeping pace with the expansion — ordinary gravitating matter invisible to every non-gravitational detector, so the forty-year null of direct searches becomes a prediction (P5 (No Particle DM)) and a confirmed particle detection would falsify the core hypothesis. Dark energy is *not* dispensed with: no arrangement of ejecta or of neighbouring universes can accelerate the expansion, and the acceleration is attributed to a constant tension of space whose value the framework does not derive. Cosmic inflation — "comic inflation" in our working vocabulary — is dispensed with as a *mechanism*: no inflaton, no slow roll; a non-singular hard-sphere bounce replaces the initial singularity and evades the Borde–Guth–Vilenkin theorem, but the horizon and flatness problems remain open and the red tilt is fitted. The rings come first (Sec. 3, H8 (Precious Rings – Dante's Purgatory)): they are phase transitions of one material, as the Earth's seismic discontinuities are, and a one-dimensional hydrodynamic toy of the rebound unloads the jammed ball from the outside in by a rarefaction wave, sorting it into three bands whose edges do not move with the kick strength. Beneath the rings is the bounce (Sec. 4, H1 (Bounce – The Big Pool Bang)): a jammed ensemble of Planck-mass relics ($\sim 10^{60}$–10^{61} survivors per present Hubble volume), treated as effective hard spheres,[1] rebounds under a loop-quantum-cosmology-type modified Friedmann equation whose critical density is conjecturally the jamming density; toy N-body simulations with a turbulent kick distribution eject $88\% \pm 4\%$ of the mass with clean velocity sorting — the mass of the dark rings. The hot Big Bang is Kernel 0,0,0 (Sec. 5): the survivor fraction is a consistency condition the merge-then-evaporate channel must meet, not a derivation, and the geometric merger cross-section misses it by up to 28 orders of magnitude, the leading open problem. The kernel interior is *assumed* homogeneous FLRW with its edge beyond the horizon; the system of rings is a finite centred cloud that still has *no bounce mechanism* ($\sim 10^{40}$ inside its own Schwarzschild radius). Coasting expansion is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$), a relativistic LTB lightcone through the ejecta decelerates ($q_0^{\text{eff}} = +0.16$), and a kinetic-Sunyaev–Zel'dovich gate caps any ejection profile at $\sim 1\%$ of the observed dark energy, so we adopt the framework *plus a constant tension of space* ($w = -1$; Sec. 6). Inside a homogeneous kernel a spherical mantle and shell exert nothing, so our position, H5 (Our Position – Where Is Nemo; Sec. 7), is ΛCDM's own null — the structure-subtracted off-centre dipole stays inside the ≈ 2.5 Mpc mock floor, as it does — while the kernel is flat to a few percent in primordial temperature and dark-to-baryon ratio, which the model's own radial gradient must respect (P12 (Kernel Gradient)). A predicted Cold Spot hot ring at 5ř–7ř

[1] Code, figures, build instructions and download links for all datasets used: https://github.com/AIMFIRST-VN/big001

is absent in Planck 2018 data; a neighbouring universe beyond our particle horizon can imprint only an initial-condition (Sachs–Wolfe) potential, which for the Cold Spot fixes a source mass $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex) 1–2.5 Gly beyond last scattering with an induced local velocity $\lesssim 1$ km s^{-1}. We organize the claims into eight hypotheses — H8 (Precious Rings – Dante's Purgatory), H1 (Bounce – The Big Pool Bang), H5 (Our Position – Where Is Nemo), H2 (Vortex – The Cold Eye of Sauron), H3 (Neighbour – Big Pool Bang Multiverse 2), H4 (Population – The Dark Comic Background), H6 (Neighbour Location) and H7 (Pool Model) — each with explicit falsifiers collected in one table, catalogue twelve predictions (P1 (Polarization Ring)–P12 (Kernel Gradient)) with the instruments required, and state the unsolved problems: the merger cross-section gap, baryogenesis, the assumed homogeneity of the kernel, the causal structure of the bounce, the relativistic formulation of ejection, the horizon and flatness problems, and the expansion history.

1. Introduction: Three Questions

The picture is an apple — the Big Apple. One kind of ball, a Planck-mass relic, fills a jammed sphere; the sphere rebounds, and the rebound sorts the balls into concentric rings by one number each, its kick relative to the escape speed. The rings are not made of different stuff; they are the same stuff under different mechanics — jammed, then collisional, then collisionless, then a wall — as the Earth's seismic discontinuities are one rock under different pressure. An apple has three distinct tissues, and that is what the hydrodynamic toy of Sec. 3 produced: the core with its seeds is Kernel 0,0,0, the innermost ring, where the balls that fell back collide and grind one another into light, leaving the surviving relics as seeds in it — the only ring with light, and where we live; the flesh is the dark mantle; and the thin skin, which comes off first, is the atmosphere that the unloading wave releases first and fastest, the dark shell of sorted ejecta. Beyond the skin is a boundary, and beyond that other bangs may lie. The paper is organized around three questions about this apple.

1. **How does the model work?** A first-order vacuum crystallization jams into a hard-sphere state; a loop-quantum-cosmology-type bounce rebounds it; the rebound sorts the relics by speed into rings (Secs. 3–4): H8 (Precious Rings – Dante's Purgatory) for the rings, H1 (Bounce – The Big Pool Bang) for the bounce beneath them. The relics that fell back form Kernel 0,0,0, where collisions grind relic mass into light until the mergers stop keeping pace with the expansion: that light is the hot Big Bang, and the cold trace of survivors is our dark matter, with no new particle (Sec. 5). The $88\% \pm 4\%$ that the toy rebound ejects are the dark rings outside. The rebound kinetics do *not* reproduce cosmic acceleration, so the adopted expansion history is the framework plus a constant tension of space whose value is not derived (Sec. 6); and the conversion calculation that must reproduce the survivor fraction has not been done (Sec. 13).

2. **Where are we?** In Kernel 0,0,0, at or near its centre, says H8 (Precious Rings – Dante's Purgatory); H5 (Our Position – Where Is Nemo; Sec. 7) tests that placement. Inside a homogeneous kernel a spherical mantle and shell exert nothing, so the placement predicts exactly what ΛCDM predicts: a displaced-centre null and a bulk flow explained by local attractors. Our current answer: the displaced-centre test is null to within a few Mpc, and the hemispherical asymmetry is better assigned to an external neighbour than to our own offset. A second handle, the radial gradient of the grinding (P12 (Kernel Gradient); Sec. 5.4), is flat to a few percent over the observable volume.

3. **Where is the second Big Bang?** Beyond the boundary ring. A neighbouring Big Pool Bang beyond our particle horizon can imprint the last-scattering surface only through the potential it contributed to the initial conditions — a Sachs–Wolfe spot with a low-multipole

tail. The Cold Spot at $(l, b) = (203ř, -56ř)$ is the candidate, treated under H2 (Vortex – The Cold Eye of Sauron), H3 (Neighbour – Big Pool Bang Multiverse 2) and H6 (Neighbour Location) in Secs. 8–10: direction known to $\pm 3ř$, comoving distance 1–2.5 Gly beyond the last-scattering surface, mass $M_2 \approx 10^{48}$ kg to within $+0.6/-0.4$ dex, and a curl-mode test, P8 (Curl Test), pre-registered. The eight extreme spots of the Planck map are the candidate list for a population of such neighbours, H4 (Population – The Dark Comic Background) and H7 (Pool Model), treated in Sec. 11.

The paper answers the questions in that order: Sec. 2 the hard-sphere ansatz and relic counts; Sec. 3 the rings, their acoustic structure and a hydrodynamic toy of the rebound; Sec. 4 the bounce and velocity sorting; Sec. 5 Kernel 0,0,0 as the hot Big Bang, the consistency checks and the kernel's radial structure; Sec. 6 the geometry by ring and the expansion history; Sec. 7 our position; Secs. 8–11 the Cold Spot, the neighbour, its location and the population; Sec. 12 the twelve predictions and the falsification table; Sec. 13 open problems. Appendix A records exploratory nulls, Appendix B a lattice check, Appendix C the kinematic-mimicry tests and the $w(z)$ drift, Appendix D the N-body ejection details.

2. Planck-Mass Relics as Effective Hard Spheres

Standard cosmology relies on scalar-field inflation to explain why the observable universe is flat, isotropic, and thermally uniform, at the cost of fine-tuning and of singularity theorems (Borde, Guth & Vilenkin 2003). The **Big Pool Bangs Theory** is a mechanical alternative grounded in first-order phase-transition dynamics. Space begins in a false vacuum with $\rho_{\text{vac}} \sim \rho_P$. Quantum fluctuations trigger a **vacuum crystallization** transition which nucleates at many points: bubbles of converted phase grow, collide, and merge, and the "bang" is their percolation into a single jammed state.

The hard-sphere ansatz. For a particle of mass m the reduced Compton wavelength $\hbar/mc$ and the Schwarzschild radius $2Gm/c^2$ coincide at $m \approx m_P$, where both equal $\sim \ell_P = 1.616 \times 10^{-35}$ m, the regime in which neither quantum field theory nor classical general relativity is separately valid. **We adopt, as a modeling ansatz rather than a derivation, that Planck relics possess a minimum spatial extent $\sim \ell_P$ that no interaction can compress**, and treat them as effective incompressible hard spheres. The ansatz is judged by the phenomenology it generates (Secs. 3–11); its microphysical justification is an open problem (Sec. 13).

Relic counts. Because quantum geometry prohibits localized infinite densities, the energy crystallizes into irreducible **Planck-mass relics** ($m_P \approx 21.8\,\mu$g). We count *survivors, per present-day Hubble volume, today*: $N \approx 10^{60}$–10^{61} relics, of mass $M_1 = N m_P \approx 2.2 \times 10^{52}$ kg at the lower end (order of magnitude; the matter mass of the *observable* universe is $\sim 1.5 \times 10^{53}$ kg). These are survivor numbers, not totals. Under H8 (Precious Rings – Dante's Purgatory) the survivors are the fraction $f = T_{\text{eq}}/T_{\text{conv}}$ of the relics that were present in the comoving Hubble volume at the bounce (Sec. 5), so the number at the bounce was $N/f \approx 10^{88}$ for Planck-scale conversion (fewer for lower T_{conv}), and the comoving region that becomes one present Hubble volume spanned, at the bounce density, $\sim 3 \times 10^{29}$ causal (Planck-size) patches per dimension. In Geometry A (Sec. 4) the jammed patch is super-horizon, so its total number and mass are unbounded by observation; in Geometry B they are the mass of a finite cloud. The relics saturate local volume up to the random close-packing limit $\eta_{\text{rcp}} \approx 0.64$, and with $\rho_P = m_P/\ell_P^3 = 5.2 \times 10^{96}$ kg m^{-3} the jamming density is $\rho_c = \eta_{\text{rcp}} \rho_P$. The Schwarzschild radius of even the survivor mass is 3.3×10^{25} m, forty orders larger than its jammed size ($\sim 10^{-15}$ m), so a finite ball of this mass and density would be a black hole, not a bouncing patch. The bounce of Sec. 4 is meaningful only if the jammed patch is the whole (or a super-horizon) universe described by a homogeneous FLRW-type metric, for which no exterior

Schwarzschild geometry exists (Geometry A); the finite-cloud reading (Geometry B) must confront this trapped-surface problem directly. We use ρ_c, not a jamming radius, as the physical parameter.

Crystallization timeline. Bubble walls propagate at $\leq c$, so percolating the $\sim 3 \times 10^{29}$ causal patches per dimension takes at least $3 \times 10^{29}\, t_P \approx 1.6 \times 10^{-14}$ s divided by the number of nucleation sites per crossing length; any nucleation density sparse enough to give nontrivial merger statistics implies an assembly time many orders above t_P, whose consistency with trapped-surface avoidance (Sec. 4) is open.

Nucleation as the origin of the kick spectrum. The randomness of the merger history is inherited by the jammed core as internal turbulence, a candidate source for the stochastic kick spectrum that Sec. 4 otherwise requires as an input — provided nucleation-era velocity structure survives the jammed epoch rather than thermalizing. A Poisson nucleation-and-merger simulation recovers a Maxwellian kick distribution (speed coefficient of variation 0.41–0.42), largely a central-limit consequence; the nontrivial statistic is the coherent-to-dispersive ratio $\sigma_{\rm rel}$. Poisson nucleation is turbulence-dominated ($\sigma/\langle v_r \rangle \approx 5$–$8$); a *cascade* model — a single seed bubble whose wall catalyzes further nucleation, halted by a jamming feedback at $\eta \approx 0.64$ — yields $\sigma_{\rm rel} = 0.65$–0.78, robust across a $4\times$ range in trigger distance. Three caveats gate the chain: the cascade's heavier-tailed kick field has not been fed through the N-body pipeline; the model retains structural choices (trigger distance, nucleation lag, wall-area weighting, the arrest $\eta = 0.64$); and the wall-momentum efficiency is assumed universal, whereas it is model-dependent and typically small (Espinosa et al. 2010). Bubble collisions in any first-order transition also source a stochastic gravitational-wave background peaked at the transition scale; this is a generic prediction of H1 (Bounce – The Big Pool Bang), listed as P3 (Transition GW Background), not of any neighbour.

3. H8 (Precious Rings – Dante's Purgatory): The Big Apple

Motivation. A single homogeneous bounce cannot be both a jammed-relic rebound whose ejecta are the dark matter and the source of a radiation era that leaves cold relics at matter–radiation equality: if every relic shares one history, the ejecta are hot at equality for any ejection Lorentz factor (Sec. 5). H8 (Precious Rings – Dante's Purgatory) resolves the conflict by *location.* The relics are identical, but they do not share one history: the kick each receives relative to the local escape speed, $f = v/v_{\rm esc}$ — the one number the rebound assigns to every relic (Sec. 4, Fig. 2) — sorts them into concentric rings, and the mechanics differ from ring to ring because the density, the collision rate and the binding differ. Dante's Purgatory is the picture: one mountain, one kind of soul, and a different law on every terrace. The Big Apple is the shape: a core with seeds, a flesh, and a skin that comes off first. Table 1 is the map.

Table 1. H8 (Precious Rings – Dante's Purgatory): the rings, their mechanics, and our handle on each.

Ring	Who is there	Mechanics	State	Becomes	Our handle
Kernel 0,0,0 (the core with its seeds)	Relics that were not released by the unloading wave (f well below 1)	Collisional: mergers, evaporation, thermalization, radiation-fluid dynamics	Light plus a trace of surviving relics	The hot Big Bang we live in: CMB, helium, baryons, and our cold dark matter as the survivors of the grinding	Everything we observe
Mantle (the flesh)	Relics that rose and fell back late, or are marginally bound ($f \approx 1$)	Gravitational, weakly collisional, mixed by many orbits	Cold relic gas, no light	A dark buffer with no memory of the initial geometry	Sets whether Kernel 0,0,0 has an edge, and how an H5 (Our Position – Where Is Nemo) offset would show
Shell (the skin)	The fast ejecta (the 88%, $f > 1$), released first	Collisionless free streaming, fastest furthest	Dark, locally velocity-sorted	The dark rings around us; the only ring that keeps the initial geometry and spin	Beyond our horizon except through initial-condition imprints
Boundary and beyond	Walls to neighbouring bangs	Vacuum-bubble physics (Coleman & De Luccia 1980)	Not matter	Cold Spot imprint, low-multipole tail	H3 (Neighbour – Big Pool Bang Multiverse 2), H6 (Neighbour Location), P8 (Curl Test), P9 (Low-ℓ Tail)

We live in Kernel 0,0,0. The hot Big Bang is Kernel 0,0,0: the ring in which collisions grind relic mass into light and leave a cold trace of survivors, our dark matter (Sec. 5). The 88% $\pm$ 4% ejected fraction of the toy rebound (Sec. 4) is the mass of the *dark rings around Kernel 0,0,0* — the shell — not the matter in our galaxies; inside Kernel 0,0,0 the dark-matter-to-baryon ratio, 5.4, is set by the survivor fraction together with baryogenesis, which remains open (Sec. 13). P5 (No Particle DM) is unchanged — the survivors are the same objects as the ejecta.

Observers live in kernels. Only Kernel 0,0,0 has light. An observer in the shell would see no

CMB, no stars, and no thermal history at all — Dante inverted: we are in the innermost ring, the only one with light, and that is why we are here to ask.

The coherence gradient. Appendix A records, as an untested hypothesis, that only the earliest and fastest ejecta preserve geometric correlations while everything slower is randomized by violent relaxation. H8 (Precious Rings – Dante's Purgatory) makes that its statement: free streaming preserves the initial geometry and spin in the shell; collisions in Kernel 0,0,0 and orbit mixing in the mantle erase them. The shell is the only ring that remembers how the bang began, and it is the ring we cannot see.

Geometry by ring. Two geometric readings of the rebound are defined in Sec. 4: Geometry A (homogeneous FLRW, no centre, no exterior) and Geometry B (a finite centred cloud whose radial ordering survives). H8 (Precious Rings – Dante's Purgatory) assigns them by location, once: the kernel interior is *assumed* homogeneous FLRW with its edge beyond the horizon (this is an assumption, not a derivation — Sec. 13, item 4(c)); the whole system of rings is Geometry B. Every observation we make is made inside Kernel 0,0,0, so the comparisons of Sec. 6 test the assumed Geometry A, and H5 (Our Position – Where Is Nemo; Sec. 7) tests our placement at or near the kernel's centre.

3.1 Acoustic structure of the rings

The rings are phase transitions of one material. The Earth's 410 km and 660 km discontinuities and its core–mantle boundary are changes of phase of one rock under pressure, and seismology maps them because sound crosses them at different speeds and reflects at the contrast. The Big Apple has the same structure with one material: jammed (the pre-bounce state, pressure diverging at $\eta = 0.64$), then a collisional gas (Kernel 0,0,0, where the relics are so dense that they merge and the merger products evaporate), then a collisionless gas (the mantle and shell), then a wall (the boundary). Each ring has its own sound speed: $c/\sqrt{3}$ in the kernel once conversion has made it a radiation fluid; the relic velocity dispersion in the mantle, where the only restoring force is gravity. In the shell there is no restoring force at all — free-streaming relics carry no sound — so the shell is acoustically dead. At the boundary the restoring force is the wall tension.

The consequence is the same as in seismology: waves from the kernel *reflect* at the kernel–mantle edge, where a radiation fluid meets a pressureless gas and the impedance contrast is total. The kernel's acoustic history is therefore confined to the kernel — the toy of Sec. 3.2 shows the reflected waves recompressing the core repeatedly — and if the kernel's edge lay inside our horizon, primordial acoustic waves that reached it before last scattering would have returned and imprinted a ring-shaped echo in the CMB at the angular scale of the edge, a feature with a preferred centre and radius unlike the statistically isotropic acoustic peaks. None is seen. Either the edge is beyond the horizon, as H8 (Precious Rings – Dante's Purgatory) assumes, or it has no impedance contrast and is not an edge in the acoustic sense; a detected centred echo would falsify the assumption, and we list it as falsifier (v) below.

3.2 A hydrodynamic toy of the rebound

The N-body toy of Sec. 4 is collisionless, and the rings are defined by collisionality; we therefore also ran a fluid toy (`rings_hydro.py`): a one-dimensional spherical Lagrangian hard-sphere gas with the Carnahan–Starling pressure and a jamming divergence at $\eta = 0.64$, self-gravity, artificial viscosity, 100 mass shells, 20 dynamical times, units $G = M = R = 1$. The ball starts jammed at $\eta = 0.60$ with an outward radial kick $v = \bar{f}\, v_{\mathrm{esc}}(r)\,(r/R)$ and no per-shell kick noise.

The ball does not explode. It *unloads from the outside in*: a rarefaction wave runs inward from the surface, and each layer is released only when the wave reaches it. The atmosphere — the skin — leaves first and fastest; deeper layers leave later and slower; the innermost never leave within the run. The ring order is therefore the *original depth*. Coarse-graining the final density profile into dense and rarefied bands gives **three bands with edges at 45% and 98% of the mass, at every kick strength** from $\bar{f} = 1.2$ to 5 (Table 2) — the three tissues of the apple: a core of 45%, a flesh of 53%, a skin of 2%. The edges are set by where the rarefaction wave stalls against gravity and pressure, not by the kick, which only decides how much of each band is unbound. The core is recompressed by reflected waves — the core-boundary velocity reverses 5–9 times in 20 dynamical times — so the kernel is made not once but repeatedly.

Table 2. Hydrodynamic toy of the rebound (`rings_hydro.py`, no kick noise, 100 shells, 20 dynamical times): unbound fraction and band structure versus kick strength.

$\bar{f}$ (kick / escape speed)	Unbound fraction (shell)	Bound (kernel + mantle)	Bands	Band edges (mass fraction)
0.7	0%	100%	2	0.45
1.0	2%	98%	2	0.45
1.2	43%	57%	3	0.45, 0.99
1.5	71%	29%	3	0.45, 0.98
2.0	88%	12%	3	0.45, 0.98
3.0	97%	3%	3	0.45, 0.98
5.0	100%	0%	3	0.45, 0.98

The unbound fraction reaches 88% at $\bar{f} = 2.0$: the fluid needs a kick of *twice* the escape speed to unbind what the collisionless N-body of Sec. 4 unbinds at 0.7 times the escape speed with turbulence. The factor is the pressure: in the fluid the kick must do work against the jamming pressure and is partly returned to the core by the reflected waves; in the N-body the relics decouple at once. That is what the rings do — switch pressure on in the kernel and off in the shell — so the two toys bracket the real rebound, collisional at the centre and collisionless at the surface. Two limits are stated plainly: runs with per-shell kick noise ($\sigma_{\rm rel} > 0$) are numerically unreliable at this resolution (shell crossings, 10^5–10^6 steps, several runs ending unbound at every shell) and are not reported; and any ring structure finer than the three bands is unresolved. The bound 45% inner band is the candidate for Kernel 0,0,0 (Sec. 5); "bound" here means *not released by the unloading wave within the run*, not gravitationally bound forever.

Falsifiers. H8 (Precious Rings – Dante's Purgatory) falls to any of: (i) a detected off-centre dipole above the mock floor, with a direction stable across shells — we would not be at the centre of Kernel 0,0,0 (Sec. 7); (ii) any measured thermal velocity dispersion of dark matter above γ/R — a warm-dark-matter detection at the keV scale would falsify it, whereas the existing Lyman-α limits, $m_{\rm WDM} \gtrsim$ few keV, H8 (Precious Rings – Dante's Purgatory) passes trivially; (iii) failure of the merge-then-evaporate conversion, once computed (merger rate, remnant stability), to reproduce $f = T_{\rm eq}/T_{\rm conv}$ (Sec. 5); (iv) any evidence that the CMB region has an edge within the horizon; (v) a centred, ring-shaped acoustic echo in the CMB at a single angular scale (Sec. 3.1); (vi) a radial gradient of the primordial temperature or of the dark-to-baryon ratio with the wrong sign — hotter or more baryonic outward (P12 (Kernel Gradient), Sec. 5.4). These are collected in Table 7. The positive predictions are P5 (No Particle DM), P11 (Cold Relics) and P12 (Kernel Gradient) (Table 6).

4. H1 (Bounce – The Big Pool Bang): The Kinetic Bounce and Velocity Sorting

Why hard-sphere pressure cannot bounce the collapse. Inside a homogeneous fluid the dynamics are governed by the Raychaudhuri equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right).$$

Near random close packing the hard-sphere kinetic pressure diverges (free-volume equation of state, $P \propto nk_BT\,[1-(\eta/\eta_{\rm rcp})^{1/3}]^{-1}$), but in general relativity positive pressure *gravitates*: no classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must be sourced by quantum-geometric corrections that dominate at the jamming density $\rho_c = \eta_{\rm rcp}\,\rho_P$. The minimal object with the required property is the loop-quantum-cosmology (LQC) modified Friedmann equation,

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \eta_{\rm rcp}\,\rho_P,$$

which encodes a maximum attainable density covariantly: $H \to 0$ as $\rho \to \rho_c$. We propose the jamming limit of the relic gas as the physical origin of the ρ_c that LQC introduces axiomatically; LQC's own Immirzi-fixed value is $\rho_c \approx 0.41\,\rho_P$ versus $0.64\,\rho_P$ here — a coincidence rather than a derivation. A numerical integration (Fig. 1) confirms that the classical hard-sphere fluid overshoots the jamming density by $> 10^4$ while the modified equation bounces at $\rho = \rho_c$; the bounce is built into the equation, not confirmed by it. Whether it completes before a trapped surface forms around the still-homogeneous patch is an assumption (Sec. 13).

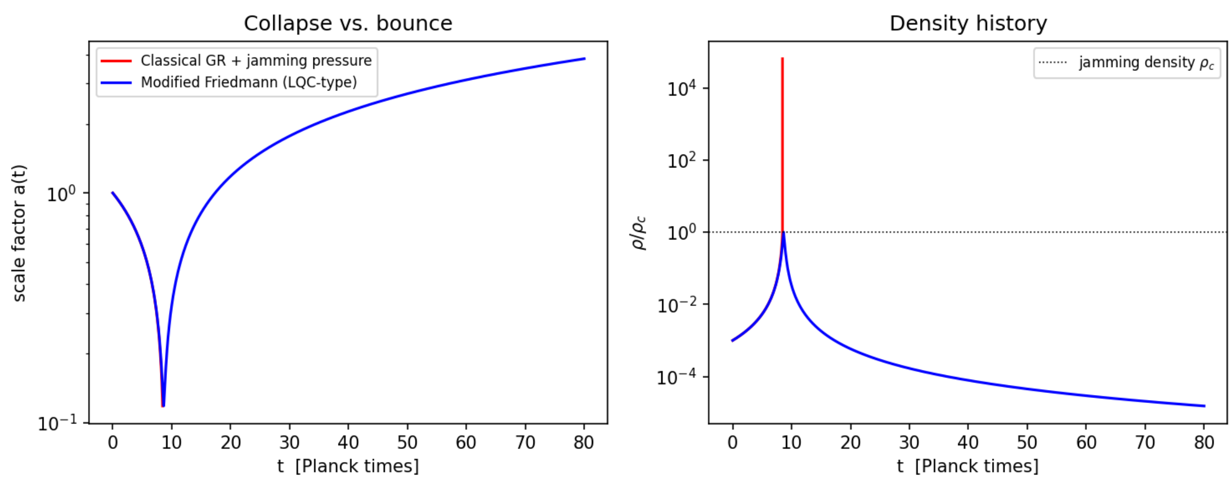

Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c.

Regime disclaimer. Everything that follows in this section is a **Newtonian analogy**: the real dynamics occur where escape velocities formally exceed c by ~ 20 orders of magnitude. We use the Newtonian model because violent-relaxation statistics are plausibly governed by dimensionless energy ratios that survive the translation; whether they do is the central open problem of Sec. 13.

The slingshot phase. Picture the jammed core as billiard balls packed to the jamming limit; the rebound is the opening break. At jamming density the relic gas is collisional, so the slingshot phase begins only after expansion dilutes the ensemble below the contact threshold and relics decouple onto ballistic trajectories; from then on the system is a self-gravitating N-body ensemble in which three-body encounters dominate relaxation — the "core collapse and halo ejection" of globular-cluster dynamics. Applied to the birth of the universe it sorts the mass into an ejected halo (the shell of Table 1: relics kicked above escape velocity, free-streaming, ordinary gravitating mass requiring no new particle) and a trapped remainder (Kernel 0,0,0 and the mantle), in which relics that fall back grind against each other; that the grinding converts relic mass to plasma while the ejecta stay stable is the conjectured channel of Sec. 5.

Velocity sorting (toy N-body; details in Appendix D). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a mean radial kick $\bar{f} v_{\rm esc}$ with a Maxwellian turbulent component of dispersion $\sigma_{\rm rel} \bar{f} v_{\rm esc}$ illustrates the ejection *mechanism*. Over a 15-seed ensemble at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, the mass of the dark rings under H8 (Precious Rings – Dante's Purgatory); the fluid toy of Sec. 3.2 reaches the same fraction at $\bar{f} = 2.0$ without turbulence, the difference being the pressure. The sorting is clean and monotonic: median speeds fall from $0.71\, v_{\rm esc}$ for the earliest ejecta to $0.28\, v_{\rm esc}$ for the latest (Fig. 2), so radial position maps onto ejection epoch; the tangential structure shows no correlated anisotropy, so any preferred axis must be seeded externally (Sec. 9). The numbers $(\bar{f}, \sigma_{\rm rel})$ remain inputs to be derived from the bounce turbulence spectrum; the ejection fraction, thresholds and kinematic statistics are in Appendix D.

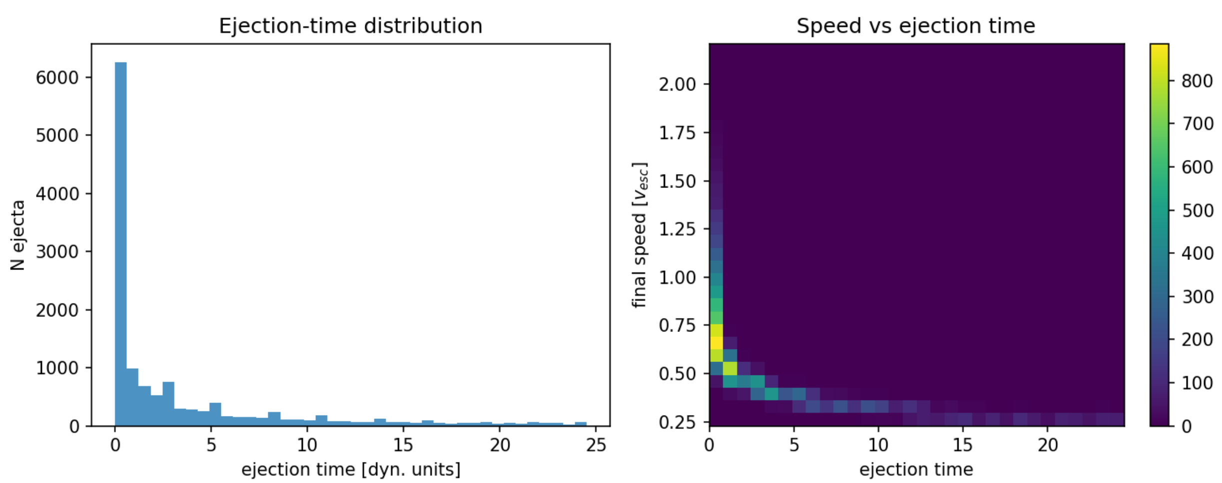

Figure 2: Fig. 2 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

Two geometries (definitions). How the sub-horizon toy maps onto the universe is not settled by the toy; two readings are defined here and assigned by ring in Sec. 3.

- **Geometry A (homogeneous).** The jammed patch is the whole (super-horizon) universe, there is no exterior, and ejection happens everywhere: every comoving region is simultaneously "trapped" and "halo". The radial shells of Fig. 2 translate into *local velocity-space structure* on an assumed FLRW background — the matter's initial peculiar-velocity distribution, which redshifts into the Hubble flow under the radiation era that conversion supplies (Sec. 5) — not into a geometric centre. This geometry avoids the trapped-surface problem of Sec. 2.

- **Geometry B (finite centred cloud).** The rebound is a single centred event and genuine radial velocity sorting survives: the fastest furthest. The Hubble-like law is ballistic, $v = r/t$, plus the adopted tension of Sec. 6. **Geometry B currently has no bounce mechanism**: a finite cloud of the relevant mass compressed to ρ_c is $\sim 10^{40}$ inside its own Schwarzschild radius, and the homogeneous LQC equation offers it no exit. Its principal observational burden is the *monopole* distance–redshift relation: the ballistic-only LTB lightcone (Appendix C) decelerates, and the LTB-plus-tension lightcone has not been computed (Sec. 6).

5. Kernel 0,0,0: The Hot Big Bang

Ejecta temperature and the bounce-to-equality energy budget. The ejecta leave relativistically, so they must be cold ($p/m_P c \lesssim 10^{-3}$) by matter–radiation equality or they are hot dark matter. Free-streaming momenta redshift as $p \propto 1/a$, so the question is how much expansion separates the bounce from equality, and that is fixed by the energy budget: $a_{\mathrm{eq}}/a_{\mathrm{bounce}}$ equals the ratio of radiation-like to rest-mass energy at the bounce. We evaluated this ratio for every source of radiation the framework offers (`energy_budget.py`). (i) If the radiation is the relics' own kinetic energy, radiation domination ends exactly when the relics stop being relativistic, because it is the same energy redshifting: for any ejection Lorentz factor from 2 to 10^{20} the ejecta reach equality with $p/m_P c = 1.6$–1.9, hot dark matter independent of γ. (ii) The trapped material cannot rescue this: the trapped relics are the slow ones, so they carry less energy than the ejecta; cold ejecta would require the trapped population to hold $\gtrsim 100$ times the ejecta's energy per particle. (iii) A radiation component $R_0 \gg \gamma$ (in relic rest-energy units) *not* sourced by the relics would give $p/m_P c = \gamma/R_0$ at equality — but would make the relics a trace $1/R_0$ of the energy at the bounce, with packing fraction $\lesssim 10^{-3}$, so that nothing is jammed; and the framework admits no such component, since it contains no residual vacuum and no reheating stage. Within a single homogeneous history, then, the jamming bounce and cold Planck-relic dark matter do not fit together. H8 (Precious Rings – Dante's Purgatory) resolves this by location: the relics that fell back and the relics that escaped have different histories, and the budget that matters for the CMB is that of Kernel 0,0,0 alone.

Which material is the kernel. An objection must be met first: material that is gravitationally *bound* after the rebound cannot become a near-critical expanding universe. The material that becomes our hot Big Bang must be the $E \approx 0$ material, and we define Kernel 0,0,0 accordingly: the ring whose total energy after conversion is near zero, expanding as near-critical FLRW. "Trapped" in the toys of Secs. 3.2 and 4 means *not released by the unloading wave*, not "gravitationally bound forever": the fluid toy's inner 45% band, held by pressure and reflected waves — recompressed, collisional, converting — is the candidate for Kernel 0,0,0. The 88% of the N-body and of the fluid toy at twice the escape speed are the shell; neither number describes the matter in our galaxies.

Conversion: merge then evaporate. A single Planck-mass relic is precisely the object usually taken to be the stable end state of black-hole evaporation, so the stability of isolated ejecta is the standard assumption rather than an extra one. In Kernel 0,0,0, however, relics collide and merge; a merger product of mass $2m_P$ is above the remnant scale and evaporates back to one relic in a few Planck times, radiating $\sim m_P$ into a thermal bath of Standard-Model quanta per merger. Repeated in a dense kernel this converts relic mass into plasma at the merger rate, while the dilute ejecta stay as relics. This is a conjecture in the semiclassical regime where Hawking's calculation is least trustworthy. Once the budget is written for Kernel 0,0,0 alone (`energy_budget.py`, case 4), the conflict above disappears, because the radiation and the dark matter of Kernel 0,0,0 are no longer the same relics at the same speed.

The depletion equation and the consistency condition. Conversion has no inverse process — the bath does not re-make relics — so the relic number density obeys the pure depletion equation

$$\frac{dn}{dt} + 3Hn = -\langle \sigma v \rangle\, n^2,$$

and stops when the merger rate $\Gamma = \langle \sigma v \rangle n$ falls below H. The survivors, a fraction f, are the local cold dark matter. Consistency with matter–radiation equality, $T_{\rm eq} \approx 0.8$ eV, fixes f in terms of the temperature $T_{\rm conv}$ at which conversion stops: the radiation-to-rest-mass ratio at that moment is $R = (1-f)/f$, and equality occurs when it has redshifted to unity, $a_{\rm eq}/a_{\rm conv} = R = T_{\rm conv}/T_{\rm eq}$, so

$$f = \frac{T_{\rm eq}}{T_{\rm conv}} : \qquad f = 7{\times}10^{-29}\ (\text{Planck-scale conversion}),\ 8{\times}10^{-26}\ (10^{16}\ \text{GeV}),\ 8{\times}10^{-20}\ (10^{10}\ \text{GeV}),\ 8{\times}10^{-13}\ (\text{TeV})$$

with g_{*s} corrections of order a factor 3 omitted; conversion below BBN's $\sim$10 MeV floor is excluded. This relation is the observed dark-matter-to-radiation ratio *rewritten*, and therefore a **consistency condition the merger channel must meet, not a derivation** of the dark-matter abundance. What the channel must supply is the cross-section: integrating the depletion equation through radiation domination, the survivor fraction is $f \approx 2H/(\langle \sigma v \rangle n)$ evaluated at conversion, which for relics that were jammed at the bounce gives the requirement

$$\langle \sigma v \rangle \approx \frac{M_P^2}{2\, T_{\rm eq}\, T_{\rm conv}} \approx 10^{28}\ \ell_P^2 c\ (\text{Planck-scale conversion})\ \ \text{to}\ \ 10^{44}\ \ell_P^2 c\ (\text{TeV}).$$

The geometric cross-section of a Planck-size sphere is $\sigma \sim \ell_P^2$, and with it $\Gamma/H \approx 0.6$ at the bounce, falling as $a^{-3/2}$ thereafter: two-body mergers with geometric cross-section freeze out *at once*, with $f \approx 1$ and essentially no light. Conversion can therefore only occur in the jammed contact phase, where every relic touches its neighbours and the two-body picture does not apply; and reaching $f \sim 10^{-28}$ requires $\sim$93 merge-and-evaporate generations (each halving the relic number) while the patch expands. Between the geometric cross-section and the required one there is a gap of up to 28 orders of magnitude, the leading open problem of this paper (Sec. 13, item 4(a)). The reflected-wave recompression of the fluid toy (Sec. 3.2) — the core driven back to high density 5–9 times in 20 dynamical times — is the mechanism any such calculation must include.

The survivors are cold in every case: a relic that leaves the collisional phase with momentum $\gamma\, m_P c$ has $p/m_P c = \gamma/R \approx \gamma f$ at equality, from 10^{-28} (Planck-scale conversion) to 10^{-12} (TeV) for $\gamma \sim 1$, and still below 10^{-3} at $\gamma = 10^3$. The light comes from the relics. We state explicitly, because the subtitle promises it, that H8 (Precious Rings – Dante's Purgatory) contains **no inflation-like phase of any kind** — no residual vacuum, no vacuum-dominated stretch, no reheating, no e-folds. The radiation era is sourced by merge-then-evaporate conversion in Kernel 0,0,0, the rings resolve the budget without any inflationary or reheating stage, and the crystallization's latent heat plays no role in the accounting.

What the bath contains. Hawking emission is democratic across species, so gravitons carry $\sim$1–2% of the bath, which appears today as $\Delta N_{\rm eff} \approx 0.01$–$0.03$ — a prediction attached to P11 (Cold Relics), just below CMB-S4's projected sensitivity. Two costs are stated. The bath is CP-symmetric, so baryogenesis must be *imported* and the local dark-matter-to-baryon ratio is not derived (Sec. 13, item 4(b)). And without inflation a GUT-scale transition in the bath would reintroduce the monopole problem, so either the bath starts below the GUT scale or the framework inherits that problem too.

5.1 Horizon problem — open

At the bounce the causal patch is of Planck size, and the region that becomes one present-day Hubble volume comprises $\sim N/f \approx 10^{88}$ such patches (of order one per relic present at the bounce; Sec. 2). Nothing in the bounce equilibrates them within a few Planck times, so homogeneity across causal patches is an assumed initial condition — in Geometry A over a super-horizon, unbounded patch; in Geometry B over the finite cloud. A long contracting phase before the bounce could in principle enlarge the causal patch, but the collapse is not modelled here.

5.2 Flatness — not addressed

In the turbulent regime the N-body simulations favor ($\bar{f} = 0.7$, $\sigma_{\rm rel} = 1.0$) the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\rm rel}^2)v_{\rm esc}^2 \approx v_{\rm esc}^2$, but this does *not* make the total energy vanish: for a uniform sphere $U = -3GM^2/5R$ and $v_{\rm esc}^2 = 2GM/R$, so $K \approx 1.7\,|U|$ (or $K \approx 2\,|U|$ with the kick scaled to the local escape speed), and $E = 0$ would require $\bar{f}^2(1 + \sigma_{\rm rel}^2) \approx 0.3$–$0.6$ depending on convention. The simulated systems are unbound, $E > 0$. The energy budget of the rebound says nothing about spatial flatness; flatness is not addressed by this framework, and the near-zero energy of Kernel 0,0,0 above is a definition of the kernel, not a derivation of flatness.

5.3 Spectral tilt — fitted

One may parametrize the scalar index by a turbulent dissipation ratio ν_t (eddy turnover time to expansion time) as $n_s - 1 = -\nu_t/3$; Planck's $n_s = 0.965 \pm 0.004$ then gives $\nu_t \approx 0.105$. Both the relation and the value are fitted inputs, and showing that a Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a nearly scale-invariant curvature spectrum is an open problem (Sec. 13).

5.4 Radial structure of the kernel and P12 (Kernel Gradient)

Grinding is more complete where the density is higher, and in a centred rebound the density is highest at the centre. Before any assumption of homogeneity, then, the model predicts a radial structure inside Kernel 0,0,0: primordial temperature *decreasing* with distance from the centre (less relic mass converted to light) and dark-to-baryon ratio *increasing* outward (more survivors per unit light). This is P12 (Kernel Gradient), a prediction with a sign. Three data sets bound it. (a) The CMB temperature–redshift relation, $T(z) = T_0(1 + z)^{1-\beta}$, measured with Sunyaev–Zel'dovich clusters and molecular absorption lines to $z \approx 3$, gives β consistent with zero at the 1–3% level. (b) The dark-matter-to-baryon ratio at last scattering (CMB acoustic peaks) and the local ratio (cluster baryon fractions, BAO) agree to a few percent. (c) The CMB is isotropic to 10^{-5}: a purely radial gradient produces no anisotropy for an observer *at* the centre, and an observer offset by d sees a gradient g as an intrinsic dipole $g \cdot d$, which the kinematic-dipole test (Doppler modulation and aberration) bounds near 10^{-4}. The kernel is therefore flat, in temperature and composition, to a few percent over the observable volume: whatever gradient the grinding produces must be saturated inside our horizon — conversion complete to the same f everywhere we can see — and the conversion calculation of item 4(a) must reproduce that saturation, not only the value of f. The prediction is falsifiable by its sign: a detected trend of the *opposite* sign — hotter, or more baryonic, with distance from us — would falsify H8 (Precious Rings – Dante's Purgatory), because no arrangement of grinding by depth produces it.

6. Geometry by Ring and the Expansion History

Geometry A and Geometry B are defined in Sec. 4 and assigned by ring in Sec. 3: the kernel interior is assumed homogeneous FLRW with its edge beyond the horizon, and the system of rings is Geometry B. This section states what the expansion history is under that assignment, and why the framework needs a constant tension.

Coasting is excluded. The rebound kinetics alone imply a coasting expansion ($a \propto t$), and this is excluded. Against 1371 Hubble-flow SNe Ia from Pantheon+ (Scolnic et al. 2022; Brout et al. 2022; diagonal errors, offset marginalized), coasting yields $\chi^2 = 674$ vs 590 for flat ΛCDM ($\Delta\chi^2 \approx 84$), though far better than Einstein–de Sitter ($\chi^2 = 1170$; Fig. 3); on DES-SN5YR (DES Collaboration 2024; Dovekie recalibration, Popovic et al. 2025; 1820 SNe, full covariance) the figures are 1632, 1763 ($\Delta\chi^2 \approx 131$), and 2566. Whether the *distribution* of the ejecta could mimic acceleration for an interior observer is tested in Appendix C: a relativistic Lemaître–Tolman–Bondi lightcone through the ejecta shells decelerates ($q_0^{\mathrm{eff}} = +0.16$); no smooth, bimodal, or lumpy profile accelerates; a blast-wave cavity is the giant-void geometry, and the kinetic Sunyaev–Zel'dovich limit on coherent cluster flows caps any cavity's contribution at $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required; and a toy effective $w(z)$ from the shells drifts a few percent near $-1/3$ in the direction *opposite* to the DES+DESI $w_0 w_a$ fit. **No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy.**

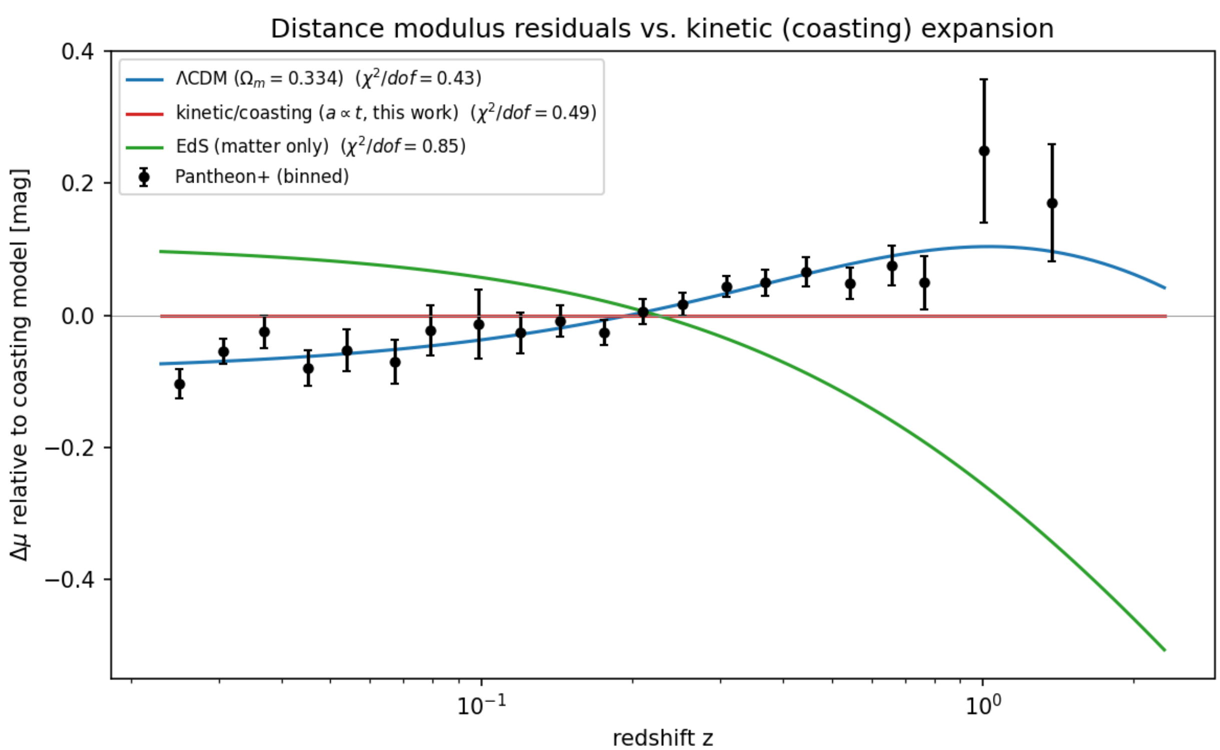

Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

The adopted expansion history: the Big Pool Bang framework plus a constant tension.

The failures above are failures of one idea only: that the *kinetics* of the rebound could substitute for dark energy. We therefore separate the two sectors: the framework is responsible for the origin and velocity structure of the matter; the acceleration is attributed to a constant tension of space, $\rho_\Lambda c^2 = -p_\Lambda$, adopted without claiming to derive it.

Why a tension. For a comoving separation $d = a\chi$, $\ddot{d} = (\ddot{a}/a)\,d$ with $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p/c^2)$: acceleration is repulsive only for $\rho + 3p/c^2 < 0$. If space stores an energy density $u \propto a^{-n}$ when stretched, thermodynamics fixes $w = n/3 - 1$. Matter ($n = 3$) attracts; strings or curvature ($n = 2$, $w = -1/3$) coast; a domain-wall *network* ($n = 1$, $w = -2/3$; many walls per Hubble volume, Bucher & Spergel 1999) repels; a constant tension ($n = 0$, $w = -1$) repels. A single wall at ~ 46 Gly contributes no such term. Observationally, Planck+BAO+SN constrain a constant equation of state to $w = -1.03 \pm 0.03$ (Planck Collaboration 2020 VI), while the DES+DESI $w_0 w_a$ fit prefers evolving w at $\sim 3\sigma$ (Appendix C). We adopt constant $w = -1$ as the minimal assumption consistent with the first, noting that the evolving-w hint, if confirmed, is not something the kinetic sector can explain: its toy drift has the wrong sign. Neighbouring Big Pool Bangs cannot contribute: a single neighbour supplies a tidal shear and a negligible induced velocity (Sec. 10), a symmetric population supplies nothing by the shell theorem (Appendix B), and neither is a monopole. Today the tension term gives $\ddot{a}/a = +3.4 \times 10^{-36}$ s^{-2} against -0.7×10^{-36} s^{-2} from matter.

Vacuum energies of neighbours do not sum. Vacuum energy acts only where it resides, so the tensions of neighbouring bangs contribute nothing here, for the same locality reason that their masses contribute no monopole (Appendix B). If all bangs crystallized into the same vacuum, the tension is one universal constant the model does not predict; if into different vacua, the differences reside in the walls between them.

The background is assumed, not derived. With the tension included, the matter of Kernel 0,0,0 is treated as pressureless dust on a flat ΛCDM background once virialized into halos; the $w_{\rm eff} \approx -0.3$ of the shell toy (Appendix C) describes the shell *kinematics*, not the background equation of state. Inside the kernel the ejection sets the matter's initial peculiar-velocity distribution on that background, those velocities redshift into the Hubble flow under the radiation era that conversion supplies (Sec. 5), and the supernova, BAO and kSZ comparisons are satisfied because the background is assumed. Geometry B — the system of rings seen as a whole — has no such shelter: there the expansion law *is* the ballistic ordering, $v = r/t$, plus the tension. The ballistic-only LTB lightcone (Appendix C) decelerates ($q_0^{\rm eff} = +0.16$) and departs from ΛCDM by ~ 0.35 mag by $z \sim 5$; the LTB-plus-Λ lightcone has *not* been computed, and until it is, Geometry B cannot claim the agreement the kernel inherits from its assumed background — the agreement that matters, since every observation we make is made inside Kernel 0,0,0. The value of the tension is not supplied: the one framework-native candidate, a residual unconverted vacuum fraction, reproduces the 10^{-123} tuning of standard cosmology and is rejected (Appendix C), and the origin of ρ_Λ is an open problem (Sec. 13), exactly as in ΛCDM.

7. H5 (Our Position – Where Is Nemo): Where Are We Relative to Our Own Big Pool Bang?

H5 (Our Position – Where Is Nemo) was conceived as the discriminator between the two geometries of Sec. 4; under H8 (Precious Rings – Dante's Purgatory) it is a test of where the rings put us. It asks: **does any residual large-scale radial ordering survive in our Hubble volume?** Two signatures would answer yes: a displaced-centre dipole $\boldsymbol{\xi}$ in the structure-subtracted peculiar-velocity field above the mock floor defined below, or a bulk-flow amplitude that grows, rather than decays, with survey depth. We state the honest limit of the test at the outset. "Inside a homogeneous kernel" is observationally FLRW: a spherical mantle and shell exert nothing on the interior by the shell theorem, so the placement of H8 (Precious Rings – Dante's Purgatory) — we are at or near the centre of Kernel 0,0,0 — predicts exactly what ΛCDM predicts — a displaced-centre null and a bulk flow explained by local attractors — and H5 (Our Position – Where Is Nemo) cannot separate H8 (Precious Rings – Dante's Purgatory) from ΛCDM. It can only falsify the placement. H8 (Precious Rings – Dante's Purgatory) differs observationally from ΛCDM through P5 (No Particle DM), P12 (Kernel Gradient), and the edge of the kernel, which is beyond the horizon and unobservable. Results are summarized in Table 3.

Displacement from the centre. The machinery is standard (King & Ellis 1973; Alnes & Amarzguioui 2006; Asvesta et al. 2022). Attributing the full CMB dipole (370 km s^{-1}) to off-centring gives $\Delta r \approx 5$ Mpc; the residual unexplained by local gravity gives $\Delta r \lesssim 1$–2 Mpc. A displaced centre enters only through the curvature of the velocity law, $\sim H|\boldsymbol{\xi}|^2/2r$. Fitting the 2{,}112 Cosmicflows-4 groups (Tully et al. 2023) at 1.5–40 Mpc with $v(R) = hR + kR^2$ diverging from a free centre gives $|\boldsymbol{\xi}| = 6.1$ Mpc toward (118ř, −24ř) at $\Delta\chi^2 = 6595$, with shells at 40–80 and 80–150 Mpc returning centres within 10ř–15ř of it. Two controls change the picture. *Mock mask:* ten isotropic mocks on CF4's exact sky positions and distance errors manufacture false centres of $|\boldsymbol{\xi}| = 0.4$–2.5 Mpc that in the outer shells point into the *same longitude band* as the data. We define the **mock floor** as the 95th percentile of the structure-subtracted mock $|\boldsymbol{\xi}|$ distribution, ≈ 2.5 Mpc with the present ten mocks. *Structure subtraction:* removing the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015) drops the inner shell to $\Delta\chi^2 = 182$, $|\boldsymbol{\xi}| = 1.7$ Mpc, within 8ř of the CMB dipole, and the outer shells into the mock range. The 1.7 Mpc residual lies *inside* the mock distribution (0.4–2.5 Mpc) and is therefore not a detection by the criterion just stated. What makes it worth one more test is its direction, which is that of the external dipole the 2M++ reconstruction itself calibrates against the CMB frame; one re-run with that dipole left free as a nuisance vector decides whether the direction is signal or calibration. Pending that re-run, **no displaced expansion centre is claimed once mapped structure is subtracted**; the current bound is $|\boldsymbol{\xi}| \lesssim 2$ Mpc, $\sim 10^{-4}$ of the distance to last scattering.

The bulk flow. The bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023) is 428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc toward (298ř ± 5ř, −8ř ± 4ř), rising with depth, in nominal $\sim 4.8\sigma$ tension with ΛCDM (Whitford et al. 2023 argue the errors are understated). Planck's kSZ analysis (Planck Collaboration 2014 XIII) limits coherent cluster flows at Gpc depth to $\lesssim 254$ km s^{-1} at 95%; the shallower CF4 flow, sourced by mapped attractors within $200\,h^{-1}$ Mpc, does not conflict with it. What the kSZ limit disfavors is a flow of $\gtrsim 250$ km s^{-1} *persisting to Gpc depth*, the second half of P4 (Off-Centre Flow). The flow axis and the CMB dipole (264ř, +48ř) lie 63ř apart, which a single offset vector does not predict; local-attractor infall explains the flow without any offset, and the neighbour of Sec. 10 contributes $\lesssim 1$ km s^{-1}. The Cold Spot direction is 86ř from the flow axis, so the southern one-sidedness (Sec. 9) is not an offset signature either.

Table 3. H5 (Our Position – Where Is Nemo): offset and bulk-flow status.

Quantity	Estimate	Method	Status
Offset from centre $\lvert\boldsymbol{\xi}\rvert$	$\lesssim 2$ Mpc; 1.7 Mpc structure-subtracted residual, inside the mock distribution	CMB dipole cap; CF4 free-centre fit with mock-mask and 2M++ controls	No detection claimed; free-dipole re-run pending

Quantity	Estimate	Method	Status
Mock floor	≈ 2.5 Mpc: 95th percentile of the structure-subtracted mock $\lvert\boldsymbol{\xi}\rvert$ distribution (mocks give 0.4–2.5 Mpc)	Ten isotropic mocks on the CF4 mask, $\Delta\chi^2 \approx 9$–61	Defines the support threshold
Offset direction	Undetermined (CMB dipole (264ř, +48ř) vs flow (298ř, −8ř))	Dipole/flow attribution	Misaligned by 63ř
Bulk flow (P4 (Off-Centre Flow))	428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc; $\gtrsim 250$ km s^{-1} persisting to Gpc depth disfavored	Cosmicflows-4, Planck kSZ	Growth with depth untested beyond $200\,h^{-1}$ Mpc

Support and falsification, in one unit. Both criteria are stated in $\lvert\boldsymbol{\xi}\rvert$ against the mock distribution. *Support for H5 (Our Position – Where Is Nemo), i.e. Geometry B:* a structure-subtracted $\lvert\boldsymbol{\xi}\rvert$ above the 95th-percentile mock floor (≈ 2.5 Mpc for the present catalogue), obtained with the 2M++ external dipole left free, with a direction stable across shells; and/or a bulk-flow amplitude that grows with depth beyond $200\,h^{-1}$ Mpc (P4 (Off-Centre Flow)). *Falsification of Geometry B on our horizon scale:* a structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ consistent with the mock distribution in the next Cosmicflows release, with a bulk flow that decays with depth. Under H8 (Precious Rings – Dante's Purgatory) the two criteria exchange roles: support for H5 (Our Position – Where Is Nemo) falsifies the placement of us at the centre of Kernel 0,0,0, and the second outcome is what H8 (Precious Rings – Dante's Purgatory) — and ΛCDM — require. The current status is neither: $\lvert\boldsymbol{\xi}\rvert \lesssim 2$ Mpc, with the 1.7 Mpc residual inside the mock distribution and the free-dipole re-run pending.[2]

8. H2 (Vortex – The Cold Eye of Sauron): The Cold Spot and the Axis of Evil

The Eridanus Cold Spot (Vielva et al. 2004; Cruz et al. 2005) is the sky's outstanding non-Gaussian feature. On the Planck SMICA map (Planck Collaboration 2020 I, IV, VII) it is centred at galactic $(l, b) = $ (203ř, −56ř), with a decrement of $-70\,\mu$K at 5ř smoothing and a peak of $-168\,\mu$K at 2ř smoothing, and a half-maximum radius of 2.5ř; we use these values throughout.

Cyclonic vortex reading. Internal quantum turbulence in the bounce creates macro-vortices. In the **Jovian/cyclonic model** these act as thermal dams that choke local heat transport, deform under shear into elongated features aligned along a preferred rotation axis (the "Axis of Evil"), and — without solid-surface friction to dissipate angular momentum — scale up with expansion and freeze into the CMB at decoupling. The Cold Spot is read as such a frozen cyclone.

Failed prediction: the thermal ring. A cyclone diverting thermal energy around its core would produce a concentric hot ring of +20–50 μK at 5ř–7ř. The azimuthally averaged Planck 2018 SMICA

[2] *Status and retirement note (distinct from falsification).* If the next Cosmicflows release returns a structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ consistent with its own tighter mock distribution, but the bulk flow neither grows nor clearly decays, then H5 (Our Position – Where Is Nemo) has no observable signature at distance-catalogue depth and is *retired as untestable at present.* Retirement is a statement about instrument reach, not a falsification, and is kept out of Table 7 for that reason.

profile ($N_{\text{side}} = 128$) around the Cold Spot, with uncertainties from 500 random high-latitude positions (Fig. 4), shows a cold core extending to $\sim$9ř (-3.4σ at 3ř) and **no hot ring at** 5ř–7ř; a +37–52 μK excess at 13ř–15ř is $\sim$2.5σ before a trials factor of $\sim$10 radial bins. The thermal-ring prediction, the first half of P1 (Polarization Ring), failed. The polarization part survives: a frozen vortex boundary should carry a radial E-mode ring of 1–3 μK at $\sim$5ř–15ř. Planck's Q_r profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK (Appendix A), so the ring is neither detected nor excluded; the test requires CMB-S4 sensitivity.

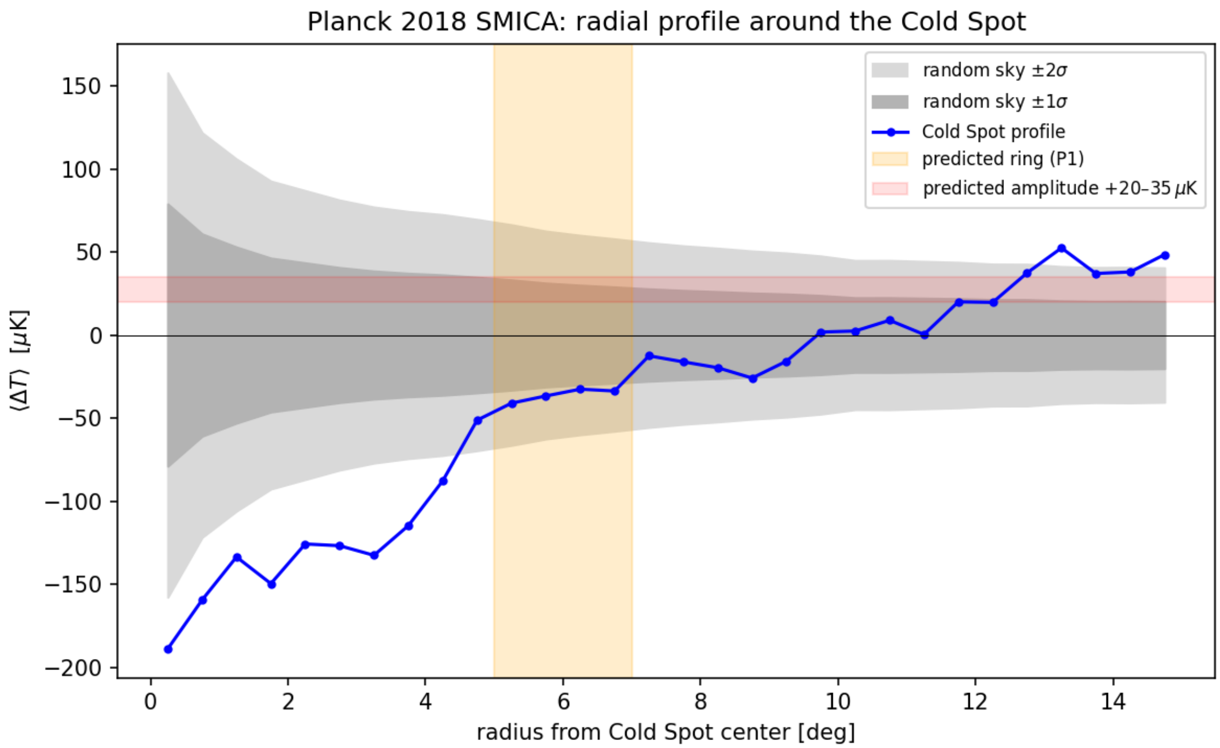

Figure 4: Fig. 4 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

Preferred axis. Our simulations find ejection isotropic within shot noise (Sec. 4), so an axis can only be supplied by an external seed, the neighbouring bang of Sec. 9. The literature's cluster of tentative ($\lesssim 3\sigma$) alignments — a supernova acceleration dipole roughly along our CMB motion (Colin et al. 2019; Sah & Rameez 2024), the quadrupole–octupole alignment with the dipole direction, coincident bulk-flow and quasar-polarization axes (Hutsemékers et al. 2005) — is motivation for H2 (Vortex – The Cold Eye of Sauron)/H3 (Neighbour – Big Pool Bang Multiverse 2), not support.

Galaxy handedness. Claimed spin-handedness dipoles (Longo 2011; Shamir 2012, 2022) lie 70ř from each other and 50ř–74ř from the axes above; we treat them as systematics-limited (Iye et al. 2021). The JWST/JADES handedness claim 8.6ř from the Cold Spot is a coincidence awaiting a control-field test (Appendix A).

Falsifier. Absence of the E-mode ring at CMB-S4 sensitivity eliminates H2 (Vortex – The Cold Eye of Sauron). A localized curl detection at the Cold Spot, P8 (Curl Test), would support the vortex reading over the neighbour reading of the same direction (Sec. 9), which is mutually exclusive with it; a calibrated curl null retires it (Table 7).

9. H3 (Neighbour – Big Pool Bang Multiverse 2) and H4 (Population – The Dark Comic Background): A Neighbouring Big Pool Bang in the Cold Spot Direction, and a Population of Neighbours

Because vacuum crystallization is stochastic across an unbounded background, our bang need not be unique. A neighbouring universe nucleating beyond our domain boundary imprints our last-scattering surface, at comoving distance $d_{\rm LS} \approx 46$ Gly (4.35×10^{26} m); a neighbour closer than that would appear in large-scale-structure tracers, which the Cold Spot sightline does not show beyond the too-shallow Eridanus supervoid.

Causal admissibility. A source at $D = (1 + \epsilon)\, d_{\rm LS}$ with $\epsilon > 0$ lies beyond our particle horizon. Nothing emitted by it since the bounce — no black holes crossing the boundary, no gravitational-wave memory, no tidal torque acting over the past Gyr — can have reached our volume. The only causally admissible imprint is one already present in the initial conditions: a potential Φ of the Grishchuk–Zel'dovich type laid down at the bounce, which the last-scattering photons sample as a Sachs–Wolfe spot with a $1/r$ tail over the whole sky at low multipoles. The neighbour reading therefore makes two kinds of prediction: the spot itself (amplitude, profile, non-rotation, polarization edge) and its low-ℓ tail, P9 (Low-ℓ Tail). Its present-day effect on our volume is a static field — an induced (monopole) velocity of the local volume and a tidal shear — fixed once the mass and distance are (Sec. 10).

The Sachs–Wolfe model. For a potential Φ present at last scattering, $\Delta T/T = \alpha\, \Phi/c^2$ (Sachs & Wolfe 1967), where the coefficient α depends on how the perturbation was laid down: $\alpha = 1/3$ for adiabatic growing modes and $\alpha \to 2$ for isocurvature-like modes in which the potential is imposed without a compensating density perturbation. An externally imposed potential lies between these limits; we adopt $\alpha = 1$ and carry the resulting factor-of-six range — asymmetric about the adopted value, $M \times 3$ at $\alpha = 1/3$ and $M/2$ at $\alpha = 2$, i.e. $+0.5/-0.3$ dex — into the mass uncertainty of Sec. 10. At the spot centre $\Phi = GM/(\epsilon\, d_{\rm LS})$, and the angular profile is

$$\Delta T(\theta) \propto \Big[\epsilon^2 + 2(1+\epsilon)(1-\cos\theta)\Big]^{-1/2},$$

so the half-maximum angle fixes the distance, $\epsilon = \theta_{1/2}/\sqrt{3}$, and the amplitude then fixes the mass, $M = (\Delta T/T)\, \epsilon\, d_{\rm LS}\, c^2/(\alpha G)$. Because $\theta_{1/2}$ is beam- and smoothing-limited at 2ř smoothing, the inferred ϵ values are upper bounds, and since $M \propto \epsilon\, \Delta T$, the mass at fixed amplitude is an upper bound too. Amplitudes and widths must be paired at the same smoothing; we use the 2ř values (ΔT, $\theta_{1/2}$) throughout. The mass function of a population of such spots is the nucleation size spectrum of the vacuum transition — the distribution that drove the cascade of Sec. 2 — so the internal (ejecta) and external (spot) samples of one process must agree, a cross-consistency requirement linking H1 (Bounce – The Big Pool Bang) and H4 (Population – The Dark Comic Background).

Southern one-sidedness. All eight most-extreme large-scale spots on the SMICA map ($|b| > 25$ř, 2ř smoothing; Appendix A) lie in the southern galactic hemisphere; in 200 Gaussian realizations through the same pipeline none placed its full top-8 in one hemisphere (mock mean 5.2 of 8). The significance to be quoted is that of the published hemispherical power asymmetry ($p \sim 1$–10%, estimator-dependent; Planck Collaboration 2020 VII), since our mocks are simplified and the frame post hoc. What this framework can say is limited: the Cold Spot and the one-sided extremes share a hemisphere, and a neighbour to the galactic south would add a few-μK potential tail on that side of the sky (Sec. 11d). Whether such an additive tail can produce the observed $\sim 7\%$ hemispherical *power* modulation, or additional $\pm 150\, \mu$K extremes, has not been computed and is not claimed; a few-μK additive term does not obviously do either. The shared hemisphere is therefore a coincidence to be explained, not a prediction of H3 (Neighbour – Big Pool Bang Multiverse 2). An isotropic swarm of neighbours (H4 (Population – The Dark Comic Background)) predicts no such one-sidedness.

H4 (Population – The Dark Comic Background). Since nucleation is stochastic, the neighbour should not be unique: a population of Big Pool Bangs at varying distances, each imprinting a distinct anomaly, P6 (Spot Population). Planck 2018 isotropy analyses (Planck Collaboration 2020 VII) find the sky consistent with Gaussian ΛCDM apart from 2–3σ large-scale anomalies and the single Cold Spot, so additional neighbours must imprint below Planck's per-spot threshold. H4 (Population – The Dark Comic Background) stands or falls on population statistics (Sec. 11), not on any single spot.

Falsifiers. H3 (Neighbour – Big Pool Bang Multiverse 2) falls to a foreground structure accounting for the full decrement, a curl detection at the Cold Spot (which transfers the direction to H2 (Vortex – The Cold Eye of Sauron)), or a P9 (Low-ℓ Tail) null at the pre-registered threshold; H4 (Population – The Dark Comic Background) to a null in sub-threshold spot statistics at LiteBIRD/CMB-S4 depth (Table 7).

10. H6 (Neighbour Location): Where Is the Second Big Pool Bang?

H6 (Neighbour Location) states the neighbour's location once, summarized in Table 4; other sections refer here.

Direction. The Cold Spot, (l, b) = (203ř, −56ř), uncertain by ∼2–3ř (the half-max radius), in the hemisphere of the eight-spot one-sidedness.

Distance. With $\theta_{1/2}$ = 2.5ř the profile of Sec. 9 gives $\epsilon = 0.025$; the eight extremes span $\epsilon = 0.025$–0.055 (Sec. 11). The source lies $\epsilon\, d_{\rm LS} \approx 1.1$ Gly comoving beyond the last-scattering surface (up to 2.5 Gly for the widest spots), at $D \approx 47$ Gly $\approx 4.5 \times 10^{26}$ m; since $\theta_{1/2}$ is smoothing-limited, ϵ is an upper bound.

Mass. With $\Delta T = -168\,\mu$K and $\theta_{1/2}$ = 2.5ř (both at 2ř smoothing), $T = 2.725$ K, $\epsilon = 0.025$, Sachs–Wolfe coefficient $\alpha = 1$ and $d_{\rm LS} = 4.35 \times 10^{26}$ m,

$$M_2 = \frac{\Delta T}{T}\,\frac{\epsilon\, d_{\rm LS}\, c^2}{\alpha G} = (6.2 \times 10^{-5})\,(1.09 \times 10^{25}\text{ m})\,(1.35 \times 10^{27}\text{ kg m}^{-1}) \approx 9 \times 10^{47}\text{ kg} \approx 10^{48}\text{ kg},$$

uncertain by $+0.6/-0.4$ dex: $+0.5/-0.3$ dex from the Sachs–Wolfe coefficient ($M \times 3$ at $\alpha = 1/3$, $M/2$ at $\alpha = 2$; Sec. 9) and ± 0.3 dex from the range of ϵ across the population (0.025–0.055), added in quadrature. Because ϵ is an upper bound, M_2 at fixed ΔT is an upper bound. The result is $\sim 4 \times 10^{-5}$ of our own per-Hubble-volume $M_1 = 2.2 \times 10^{52}$ kg, a dwarf neighbour. We use $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex) throughout.

Induced velocity of the local volume. The neighbour's present-day acceleration of our volume is $GM_2/D^2 = (6.67 \times 10^{-11})(9 \times 10^{47})/(4.5 \times 10^{26})^2 \approx 3 \times 10^{-16}$ m s^{-2}; integrated over a Hubble time (4.4×10^{17} s) this is ≈ 130 m s^{-1}, the induced (monopole) velocity of the local volume — ~ 0.1 km s^{-1}, and $\lesssim 1$ km s^{-1} after the mass uncertainty. This is negligible against the 428 ± 36 km s^{-1} Cosmicflows-4 bulk flow at $200\,h^{-1}$ Mpc (Watkins et al. 2023) toward (298ř, −8ř), 86ř from the Cold Spot. The neighbour reading therefore predicts *no* bulk-flow contribution above ~ 1 km s^{-1}, stated as P10 (Induced-Velocity Bound); the observed flow must be explained by local attractors. The genuinely tidal part is the shear: a tidal tensor $GM_2/D^3 \approx 7 \times 10^{-43}$ s$^{-2} \approx 1.5 \times 10^{-7}\, H_0^2$ (with $H_0^2 \approx 4.8 \times 10^{-36}$ s^{-2}), which integrated over a Hubble time gives a dimensionless expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$, undetectable in supernova anisotropy searches, far below the limits on anisotropic expansion (Saadeh et al. 2016) and below the low-ℓ Grishchuk–Zel'dovich (1978) limit — a consistency check, not a discriminant. Hot spots require a boundary imprint of the opposite sign (an underdense or approaching wall); the model does not specify which neighbours imprint hot.

Our position relative to it. ≈ 47 Gly comoving along the axis toward (203ř, −56ř); the internal offset of H5 (Our Position – Where Is Nemo), $\lesssim$ few Mpc (Sec. 7), is negligible by comparison.

Table 4. H6 (Neighbour Location): the neighbour's direction, distance, mass and local effects.

Quantity	Estimate	Method	Status
Direction	(203ř, −56ř) ± 3ř	Cold Spot centroid; same hemisphere as the eight extremes	Fixed by data; attribution hypothetical
Distance beyond last scattering	$\epsilon = 0.025$, ≈ 1.1 Gly comoving (upper bound)	$\epsilon = \theta_{1/2}/\sqrt{3}$ at 2ř smoothing	Smoothing-limited
Mass M_2	$\approx 10^{48}$ kg $(+0.6/-0.4$ dex; upper bound at fixed $\Delta T)$	Sachs–Wolfe, coefficient $\alpha = 1$ (1/3–2), $\Delta T = -168\,\mu$K at 2ř	Single adopted value
Induced local velocity (P10 (Induced-Velocity Bound))	$\lesssim 1$ km s^{-1}	$GM_2/D^2 \times t_H$ (monopole)	Bound; flow is local
Tidal tensor	$\sim 1.5 \times 10^{-7}\, H_0^2$ (expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$)	GM_2/D^3	Consistent, undetectable
Curl at Spot (P8 (Curl Test))	Zero (neighbour) vs excess (vortex)	Lensing curl / B-mode stack	Uncalibrated null; calibrated test pending
Low-ℓ tail (P9 (Low-ℓ Tail))	Fixed $\ell = 2, 3$ template	Correlation vs Gaussian null (Sec. 11)	Uninformative at present

What H6 (Neighbour Location) predicts. A *non-rotating* imprint (zero curl at calibrated sensitivity), a disc-shaped edge with the Aguirre & Johnson (2011) bubble-collision polarization profile, a low-ℓ tail of fixed shape (P9 (Low-ℓ Tail)), no bulk-flow contribution (P10 (Induced-Velocity Bound)), and no proper motion. It does not predict the hemispherical power asymmetry (Sec. 9). Falsifiers are in Table 7 (Sec. 12); P8 (Curl Test) is the decisive near-term test, separating the two readings of the same direction.

11. H7 (Pool Model): The Population

(a) Universe 1 is one of several. Vacuum crystallization nucleated more than once; our bang is one of several separated by domain boundaries, each with its own kernel and its own dark rings.

(b) Each neighbour imprints one spot. Under the Sachs–Wolfe model of Sec. 9, the eight most extreme large-scale spots on the SMICA map are the candidate neighbour list (Table 5), each with $\epsilon = \theta_{1/2}/\sqrt{3}$ (an upper bound, since $\theta_{1/2}$ is smoothing-limited at 2ř; amplitudes and widths are both the 2ř-smoothing values, and each implied mass carries the $+0.6/-0.4$ dex of Sec. 10):

Table 5. The eight most extreme large-scale spots of the Planck SMICA map ($|b| > 25$ř, 2ř smoothing).

Rank	(l, b)	Sign	ΔT (μK)	$\theta_{1/2}$	ϵ (upper bound)
1	(157ř, −71ř)	hot	+170	2.5ř	0.025
2	(80ř, −33ř)	cold	−170	3.5ř	0.035
3	(203ř, −56ř) (Cold Spot)	cold	−168	2.5ř	0.025
4	(155ř, −29ř)	cold	−163	5.5ř	0.055
5	(305ř, −29ř)	hot	+153	2.5ř	0.025
6	(211ř, −35ř)	cold	−153	2.5ř	0.025
7	(170ř, −47ř)	hot	+162	4.5ř	0.045
8	(184ř, −54ř)	hot	+155	4.5ř	0.045

The eight extremes require $\epsilon = 0.025$–0.055, i.e. sources 1–2.5 Gly (comoving) beyond last scattering. Under Sachs–Wolfe an overdense wall gives a *cold* spot; a hot spot requires an underdensity or an approaching (Doppler) wall, a sign mechanism the model leaves unspecified (Sec. 10). Spot 8 at (184.2ř, −54.3ř) lies 11ř from the Cold Spot centroid (203.2ř, −56.3ř) and may be part of one system; spot 7 at (170.3ř, −46.6ř) lies 22ř away and is separate. The candidate count is ≤ 8, or 7 if spot 8 is merged.

(c) Population predictions. Neighbours cluster on the sky (currently 8/8 south). Any genuine neighbour is non-rotating and should show zero curl; any relic vortex of our own bang, H2 (Vortex – The Cold Eye of Sauron), should show a curl excess under P8 (Curl Test) — the population divides on this test. The inferred neighbour mass function must match the nucleation spectrum inferred from our own ejecta. The exploratory locus, curl, lensing-convergence and tracer statistics on the eight spots are reported once, in Appendix A; none is significant.

(d) The low-multipole tail, P9 (Low-ℓ Tail). A neighbour that imprints a compact spot cannot imprint *only* a spot: the $1/r$ potential of each source extends over the whole sky at $\sim$2% of its peak (3–6 μK at 90ř, 2–4 μK at the antipode). Summing the eight tails gives a fixed $\ell = 1$–3 template with rms 4.1 μK at $\ell = 2$ and 4.5 μK at $\ell = 3$, i.e. $\sim$20% of the observed quadrupole power ($C_2 \approx 210\,\mu\mathrm{K}^2$) and $\sim$8% of the octopole — allowed, and a prediction: the template must correlate positively with the observed $\ell = 2, 3$ maps. *Pre-registered threshold.* With only 5 ($\ell = 2$) and 7 ($\ell = 3$) real modes per multipole, the correlation coefficient r between independent Gaussian patterns has $\sigma \approx 0.4$–0.5 under the null, so a single value of r is uninformative. The test is: compute r between the template and the masked Planck Commander $\ell \leq 3$ maps (Planck Collaboration 2020 IV), and the same r for 1000 Gaussian ΛCDM realizations (HEALPix synfast, same mask and pipeline); a detection requires r above the 95th percentile of the null at *both* multipoles, and a value at or below the null median at both, with the template amplitude fixed by the spots, falsifies the population reading. An unmasked SMICA comparison ($r = +0.15$ at $\ell = 2$, -0.27 at $\ell = 3$; template axes 78ř and 52ř from the observed) is uninformative at this power and foreground-contaminated.

(e) One-sided or surrounding. A *surrounding* population would populate both hemispheres; the 8/8 southern count indicates a one-sided, clustered population, and H7 (Pool Model) takes that as its picture. The symmetric lattice of Appendix B is the limiting symmetric case, whose cubic-harmonic ($\ell = 4$) expansion-rate signature is a further falsifier. The pre-registered extension is the hemispheric balance of the extremes ranked 9–28: a one-sided population predicts a persistent southern excess; a surrounding population, or a Gaussian sky with the top-8 count a fluctuation, predicts isotropy. What falsifies H7 (Pool Model) as a whole is given in Table 7 (Sec. 12).

12. Falsifiable Predictions, Instrumentation, and Falsification Criteria

Hypothesis hierarchy. H8 (Precious Rings – Dante's Purgatory; P5 (No Particle DM), P11 (Cold Relics), P12 (Kernel Gradient)) is the organizing statement and H1 (Bounce – The Big Pool Bang; P3 (Transition GW Background), P5 (No Particle DM)) the mechanism beneath it: H8 (Precious Rings – Dante's Purgatory) presupposes a rebound that sorts by speed, and H1 (Bounce – The Big Pool Bang) can stand without H8 (Precious Rings – Dante's Purgatory) only as a bounce with hot relics. H5 (Our Position – Where Is Nemo; P4 (Off-Centre Flow)) tests the placement H8 (Precious Rings – Dante's Purgatory) assigns us and cannot separate H8 (Precious Rings – Dante's Purgatory) from ΛCDM (Sec. 7). H2 (Vortex – The Cold Eye of Sauron; P1 (Polarization Ring), P2 (Cyclonic B-modes), P7 (Spin Surcharge), P8 (Curl Test)), H3 (Neighbour – Big Pool Bang Multiverse 2; P8 (Curl Test), P9 (Low-ℓ Tail)), H4 (Population – The Dark Comic Background; P6 (Spot Population)), H6 (Neighbour Location; P8 (Curl Test), P9 (Low-ℓ Tail), P10 (Induced-Velocity Bound)) and H7 (Pool Model; P6 (Spot Population), P9 (Low-ℓ Tail)) are separable from the core: any can fail without falsifying the core mechanism. They are not all mutually independent. H2 (Vortex – The Cold Eye of Sauron) and H3 (Neighbour – Big Pool Bang Multiverse 2) are *mutually exclusive* readings of the Cold Spot direction, and P8 (Curl Test) decides between them: a localized curl excess transfers the direction to H2 (Vortex – The Cold Eye of Sauron), a calibrated null to H3 (Neighbour – Big Pool Bang Multiverse 2). H6 (Neighbour Location) stands or falls with H3 (Neighbour – Big Pool Bang Multiverse 2), and H7 (Pool Model) presupposes H4 (Population – The Dark Comic Background). H8 (Precious Rings – Dante's Purgatory) and H5 (Our Position – Where Is Nemo) are linked in the opposite sense: a positive H5 (Our Position – Where Is Nemo) result (P4 (Off-Centre Flow)) falsifies the placement of us at the centre of Kernel 0,0,0. The remaining pairs can hold or fail independently.

Table 6. Predictions.

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P1 (Polarization Ring)	Cold Spot ring (H2 (Vortex – The Cold Eye of Sauron)). Thermal ring of +20–50 μK at 5ř–7ř: **failed** in Planck 2018 (Sec. 8). Surviving prediction: radial E-mode ring of 1–3 μK at 5ř–15ř	Polarization sensitivity $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ at $\ell \sim 20$–100	CMB-S4	Thermal: failed. Polarization: open
P2 (Cyclonic B-modes)	Cyclonic B-mode pattern aligned with the Axis of Evil (H2 (Vortex – The Cold Eye of Sauron))	Degree-scale B-modes, r-equivalent $\sim 10^{-3}$	CMB-S4, LiteBIRD	Open
P3 (Transition GW Background)	First-order-transition stochastic gravitational-wave background from bubble collisions in the crystallization (H1 (Bounce – The Big Pool Bang)): broken power law peaked at the transition scale	Stochastic-background spectral shape at nHz–mHz	Pulsar timing arrays, LISA	Open

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P4 (Off-Centre Flow)	Radial velocity ordering in our Hubble volume — Geometry B's own prediction (H5 (Our Position – Where Is Nemo)): a structure-subtracted, free-dipole off-centre dipole $\lvert\boldsymbol{\xi}\rvert$ above the 95th-percentile mock floor (≈ 2.5 Mpc, Sec. 7) with a direction stable across shells, and/or a bulk flow that *grows* with depth beyond $200\,h^{-1}$ Mpc. The kernel placement of H8 (Precious Rings – Dante's Purgatory), like ΛCDM, predicts a null; a flow $\gtrsim 250$ km s^{-1} persisting to Gpc depth is disfavored by the Planck kSZ limit	Next Cosmicflows release (free-dipole re-run); peculiar-velocity survey to $z \sim 1$ with $\lesssim 300$ km s^{-1} per cluster	CF4 (archival re-run); Rubin/LSST + Euclid + kSZ	$\lvert\boldsymbol{\xi}\rvert \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution; flow rising to $200\,h^{-1}$ Mpc, errors disputed
P5 (No Particle DM)	No particle dark-matter candidate (DM = Planck-relic survivors of the grinding in Kernel 0,0,0) (H1 (Bounce – The Big Pool Bang); H8 (Precious Rings – Dante's Purgatory))	Continued nulls approaching the neutrino floor	LZ, XENONnT	Ongoing

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P6 (Spot Population)	A sub-threshold population of Cold-Spot-class spots (H4 (Population – The Dark Comic Background); H7 (Pool Model)); Planck already limits any above-threshold population	Sub-threshold spot statistics; per-candidate polarization at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$; hemispheric balance of extremes 9–28	LiteBIRD, CMB-S4; Planck (archival)	Constrained; open below threshold
P7 (Spin Surcharge)	Spin surcharge (H2 (Vortex – The Cold Eye of Sauron)): filaments and clusters carry primordial angular momentum atop tidal spin, largest at high z, randomly oriented ($\omega/H_0 \lesssim 10^{-9}$; cf. the lattice-spin cap of Sec. 13); the Wang et al. (2021) filament spin is the $z \approx 0$ anchor	Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM torque mocks	DESI, Euclid, 4MOST	Open

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P8 (Curl Test)	Localized curl excess (lensing curl and B-mode variance) at the extreme-spot positions if they are relic vortices (H2 (Vortex – The Cold Eye of Sauron)); zero at the Cold Spot if it is a neighbour imprint (H3 (Neighbour – Big Pool Bang Multiverse 2); H6 (Neighbour Location)). A calibrated stacked null at the Cold Spot retires the vortex reading (Appendix A)	Planck PR4 curl reconstruction at full depth; degree-scale B-modes at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	Planck PR4 (archival), CMB-S4, LiteBIRD	Uncalibrated null; calibrated test pending
P9 (Low-ℓ Tail)	Low-multipole tail (H3 (Neighbour – Big Pool Bang Multiverse 2); H6 (Neighbour Location); H7 (Pool Model)): the fixed $\ell = 1$–3 template summed from the eight point-mass profiles must correlate positively with the observed $\ell = 2, 3$ maps, with r above the 95th percentile of 1000 Gaussian realizations at both multipoles (Sec. 11d)	Masked Commander low-ℓ maps (Planck Collaboration 2020 IV); r against the simulated null	Planck 2018/PR4 (archival)	Unmasked $r = +0.15$ ($\ell = 2$), -0.27 ($\ell = 3$): uninformative

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P10 (Induced-Velocity Bound)	Induced-velocity bound (H6 (Neighbour Location)): the neighbour's induced velocity of the local volume is $\lesssim 1$ km s^{-1} (Sec. 10), so the observed 428 km s^{-1} flow must be accounted for by local attractors	Bulk-flow residual after 2M++/CF4 reconstruction $\ll 400$ km s^{-1} unexplained	CF4 (archival); DESI peculiar velocities	Bound; consistent
P11 (Cold Relics)	Cold relics (H8 (Precious Rings – Dante's Purgatory)): a consistency statement, not a discriminant — any dark matter above the keV scale produced non-relativistically is cold, and the survivors have $p/m_P c \approx \gamma T_{\rm eq}/T_{\rm conv} < 10^{-6}$ at equality (Sec. 5), so there is no free-streaming cutoff; P11 (Cold Relics) constrains warm dark matter, not ΛCDM. Added: the democratic Hawking bath puts $\sim$1–2% of its energy in gravitons, $\Delta N_{\rm eff} \approx 0.01$–$0.03$	Small-scale matter power: no cutoff at any k ($m_{\rm WDM} \to \infty$); $N_{\rm eff}$ to ± 0.03	Lyman-α forest, strong-lens substructure, 21-cm; CMB-S4	Consistent ($m_{\rm WDM} \gtrsim$ few keV); $\Delta N_{\rm eff}$ below current sensitivity

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P12 (Kernel Gradient)	Kernel gradient (H8 (Precious Rings – Dante's Purgatory)): primordial temperature decreasing and dark-to-baryon ratio increasing with distance from the kernel's centre, saturated over the observable volume (Sec. 5.4); a trend of the *opposite* sign falsifies H8 (Precious Rings – Dante's Purgatory)	$T(z) = T_0(1+z)^{1-\beta}$ with β at the percent level; dark-to-baryon ratio at last scattering vs local; CMB intrinsic dipole $\lesssim 10^{-4}$	SZ temperature measurements (Planck, SPT, ACT), molecular absorption; high-z cluster baryon fractions (eROSITA, Euclid); CMB dipole aberration	Flat to a few percent (β consistent with zero at 1–3% to $z \approx 3$); consistent

Table 7. Falsification criteria. This is the single reference for what each hypothesis needs and what removes it; Secs. 11, 13 and the Epilogue refer here.

Hypothesis	What supports it	What falsifies it	Current status
H8 (Precious Rings – Dante's Purgatory) rings: Kernel 0,0,0 with cold relic survivors, dark rings outside	Displaced-centre null with a decaying bulk flow; no free-streaming cutoff (P11 (Cold Relics)); particle-DM nulls (P5 (No Particle DM)); a kernel flat to a few percent, with any detected gradient of the predicted sign (P12 (Kernel Gradient)); a conversion calculation that reproduces $f = T_{\mathrm{eq}}/T_{\mathrm{conv}}$	Any of: an off-centre dipole above the mock floor, stable across shells; a dark-matter thermal velocity dispersion above γ/R (keV-scale warm dark matter); failure of the conversion calculation to reproduce f; an edge of the CMB region within the horizon; a centred ring-shaped acoustic echo (Sec. 3.1); a kernel gradient of the wrong sign (P12 (Kernel Gradient))	Open; displaced-centre null, Lyman-α limits and $T(z)$ flatness consistent; conversion calculation not done, cross-section gap of up to 28 orders

Hypothesis	What supports it	What falsifies it	Current status
H1 (Bounce – The Big Pool Bang) bounce and velocity sorting	Clean velocity sorting in both toys; particle-DM nulls (P5 (No Particle DM)); a transition-shaped GW background (P3 (Transition GW Background))	A particle dark-matter detection; an ejecta velocity spectrum that cannot yield a cold matter component by equality; a derived ρ_c differing from the LQC 0.41 ρ_P by more than a factor of 3	Open; relativistic formulation unresolved; ejecta temperature resolved by location under H8 (Precious Rings – Dante's Purgatory)
H5 (Our Position – Where Is Nemo) residual radial ordering (Geometry B vs A)	Structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ above the 95th-percentile mock floor (≈ 2.5 Mpc at present) with a direction stable across shells; bulk flow growing with depth (P4 (Off-Centre Flow)). Either supports Geometry B and falsifies the kernel placement	Structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ consistent with the mock distribution in the next Cosmicflows release, together with a bulk flow decaying with depth: falsifies Geometry B on our horizon scale (retirement as untestable is a separate status, Sec. 7 footnote); this outcome is what H8 (Precious Rings – Dante's Purgatory) and ΛCDM both require	Unresolved; $\lvert\boldsymbol{\xi}\rvert \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution, awaiting the free-dipole re-run
H2 (Vortex – The Cold Eye of Sauron) Cold Spot vortex	E-mode ring (P1 (Polarization Ring)), cyclonic B-modes (P2 (Cyclonic B-modes)), curl excess at the Spot (P8 (Curl Test)), spin surcharge (P7 (Spin Surcharge))	Absence of the E-mode ring at CMB-S4 sensitivity; a calibrated stacked curl null at the Cold Spot	Thermal ring failed; polarization open; curl uncalibrated
H3 (Neighbour – Big Pool Bang Multiverse 2) neighbour in the Cold Spot direction	Zero curl at the Spot (P8 (Curl Test)); positive low-ℓ tail correlation (P9 (Low-ℓ Tail)); bubble-collision polarization edge	A foreground structure accounting for the full decrement; a curl detection at the Spot; P9 (Low-ℓ Tail) at or below the null median at both multipoles	Open; P9 (Low-ℓ Tail) uninformative

Hypothesis	What supports it	What falsifies it	Current status
H4 (Population – The Dark Comic Background) population of neighbours	Sub-threshold spot population (P6 (Spot Population)); mass function matching the ejecta nucleation spectrum	Null sub-threshold spot statistics at LiteBIRD/CMB-S4 depth	Constrained above threshold; open below
H6 (Neighbour Location) location of the second bang	Non-rotating imprint (P8 (Curl Test)); low-ℓ tail (P9 (Low-ℓ Tail)); no induced-velocity contribution to the bulk flow (P10 (Induced-Velocity Bound)); Aguirre–Johnson polarization edge	Curl detection at the Spot; absence of the bubble-collision polarization profile at CMB-S4 sensitivity	Open
H7 (Pool Model) one-sided population	Southern excess persisting in extremes 9–28 (P6 (Spot Population)); positive P9 (Low-ℓ Tail); zero curl at all eight spots	All of: P9 (Low-ℓ Tail) failing at the pre-registered threshold; zero curl at calibrated sensitivity at all eight spots *and* no bubble-collision polarization profile at any at CMB-S4 depth; no CMB-independent mass counterpart in a pre-registered galaxy-density re-run. For the lattice limit: a detected $\ell = 4$ cubic expansion-rate pattern (Appendix B)	Open; 8/8 south at $p \sim 1$–10%

Under the joint nulls of H3 (Neighbour – Big Pool Bang Multiverse 2), H4 (Population – The Dark Comic Background), H6 (Neighbour Location) and H7 (Pool Model) the spot population is Gaussian and the Cold Spot a single outlier of unknown origin, while H8 (Precious Rings – Dante's Purgatory), H1 (Bounce – The Big Pool Bang) and H2 (Vortex – The Cold Eye of Sauron) stand and H5 (Our Position – Where Is Nemo) continues to test whether we sit at the centre of Kernel 0,0,0 as H8 (Precious Rings – Dante's Purgatory) requires. The polarization ring and calibrated curl test await Simons Observatory / CMB-S4 (a 2030s timeline); P9 (Low-ℓ Tail), P10 (Induced-Velocity Bound) and P12 (Kernel Gradient) are archival.

13. Limitations and Open Problems

1. **Hard-sphere ansatz.** Treating Planck relics as incompressible spheres is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a quantum-gravity result.
2. **Bounce mechanism and horizon structure.** The rebound requires the LQC-form modified Friedmann equation (Bojowald 2001; Ashtekar & Singh 2011); $\rho_c = \eta_{\rm rcp}\rho_P$ is a conjecture. The bounce is meaningful without qualification only for a whole or super-horizon patch (Geometry A). The finite-cloud Geometry B *currently has no bounce mechanism*: a finite cloud of the relevant mass is $\sim 10^{40}$ inside its Schwarzschild radius, the homogeneous LQC equation offers it no exit, and no trapped-surface timing analysis (which must also accommodate the assembly timescale $\gtrsim 3 \times 10^{29}\, t_P$ per Hubble volume) has been done.
3. **Spectral index.** ν_t and the relation $n_s - 1 = -\nu_t/3$ are fitted inputs; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.
4. **The kernel.** H8 (Precious Rings – Dante's Purgatory) resolves the bounce-to-equality conflict of Sec. 5 by location, at a price that is stated, not paid. **4(a) The merger cross-section gap (up to 28 orders).** The survivor fraction $f = T_{\rm eq}/T_{\rm conv}$ is a consistency condition, not a derivation, and meeting it requires $\langle \sigma v\rangle \approx 10^{28}$–$10^{44}\, \ell_P^2 c$, against a geometric cross-section $\sim \ell_P^2$ with which mergers freeze out at the bounce with $f \approx 1$ and no light. Conversion must happen in the jammed contact phase through ~ 93 merge-and-evaporate generations, with the reflected-wave recompression of Sec. 3.2 included; that calculation has not been done. **4(b) Baryogenesis.** The bath is CP-symmetric, so the baryon asymmetry must be imported and the local dark-matter-to-baryon ratio of 5.4 is not derived; a GUT-scale transition in the bath would reintroduce the monopole problem without inflation. **4(c) Kernel homogeneity assumed.** That the kernel interior is homogeneous FLRW with its edge beyond the horizon is an assumption (Sec. 3), tested only by the displaced-centre null, the absence of an acoustic echo, and the few-percent flatness of P12 (Kernel Gradient); whether the kernel–mantle boundary is sharp is not modelled. **4(d) Relic and causal-patch counts under H8 (Precious Rings – Dante's Purgatory).** The survivor count $N \approx 10^{60}$–10^{61} per Hubble volume implies $N/f \approx 10^{88}$ relics and $\sim 3 \times 10^{29}$ causal patches per dimension in the comoving Hubble volume at the bounce for Planck-scale conversion; homogeneity across those patches is assumed. Planck-relic dark matter must also face femtolensing, capture, and abundance constraints, and the relativistic formulation of the ejection remains open.
5. **Expansion history.** No ejection profile produces apparent acceleration, the toy's $w(z)$ drift runs opposite to the DES+DESI direction (Appendix C), and the kSZ gate caps any ejecta contribution at $\sim 1\%$ of dark energy. The adopted history (Sec. 6) is the framework plus a constant tension on an assumed ΛCDM background inside the kernel; the tension's value is not predicted, and the LTB-plus-tension lightcone of Geometry B has not been computed. The model is also not yet reconciled with BBN abundances or the CMB acoustic peaks.
6. **Causal and relativistic structure.** Ejection proceeds where Newtonian escape velocity vastly exceeds c; the toys lack a relativistic formulation, prerequisite for H1 (Bounce – The Big Pool Bang) to be a physical theory rather than an analogy. How the initial-condition potential of a neighbour (Sec. 9) is laid down across a

domain boundary is not modelled.

7. **Sign of hot spots and the Sachs–Wolfe coefficient.** Sachs–Wolfe supplies cold spots from overdense walls; the hot extremes require an unspecified underdensity or Doppler mechanism (Sec. 10). The coefficient relating Φ to $\Delta T/T$ for an externally imposed potential is bracketed (1/3–2, a factor of six) rather than derived, which is the dominant term in the $+0.6/-0.4$ dex on M_2.
8. **Flatness and one-sidedness.** Spatial flatness is not addressed by the energy budget of the rebound (Sec. 5.2). Whether a neighbour's few-μK potential tail can generate the observed hemispherical power modulation has not been computed (Sec. 9).

Each item maps onto the observational program of Sec. 12 (Tables 6 and 7); the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM.

Further research: spin and an equal-mass lattice. A bang born from an isotropic bounce carries no net spin above shot noise (the per-sky ejection dipole of Appendix D matches the shot-noise expectation) and acquires spin only through tidal torque from neighbours, at second order (rate $\sim q\,GM/D^3$ for a cloud of asymmetry q). If every lattice bang had our own mass per Hubble volume, $M_1 = 2.2 \times 10^{52}$ kg, two order-of-magnitude constraints follow. First, the $1/r$ potential tail of an equal-mass neighbour keeps the CMB quadrupole below $\sim 4\,\mu$K only for spacings $D \gtrsim 10^3$ Gly ($\sim 20\,d_{\rm LS}$); at the Cold Spot distance an equal-mass neighbour would imprint a $\sim$4 K spot ($\gtrsim 28$ K for the $\gtrsim 1.5 \times 10^{53}$ kg of a Geometry-B cloud, with the safe spacing rising to $D \gtrsim 2 \times 10^3$ Gly), so the Cold Spot source is either a bang of order 10^{-5}–10^{-4} of our mass or not a bang at all. Second, because the quadrupole bound fixes the minimum spacing, the tidally induced global spin of an equal-mass lattice is *capped* at $\omega/H_0 \lesssim 10^{-8}\,(q/0.1)$ with shear $\sim 10^{-7}$: a detected global spin above the cap would *contradict* the equal-mass lattice, a detection at or below it would fix the spacing. The Saadeh et al. (2016) limit on homogeneous Bianchi VII$_h$ vorticity, $(\omega/H_0)_0 \lesssim 10^{-10}$, is derived for decaying modes whereas a tidal spin is a growing second-order mode; its CMB signature is the same quadrupole/spiral pattern, and if the bound transfers directly the spacing rises to $D \gtrsim 5 \times 10^3$ Gly. The primordial spin surcharge of P7 (Spin Surcharge) is randomly oriented and averages to zero on large scales, whereas the lattice spin is a single global axis; the two are separately bounded. A treatment of tidal-torque spin-up of a finite cloud in Geometry B is left to a separate paper.

Two roads not taken. (i) *Domain walls as dark energy.* A network of walls does accelerate expansion, but with $w = -2/3$, excluded by the measured $w = -1.03 \pm 0.03$; a single wall beyond the horizon exerts no force on the interior (Sec. 6, Appendix B). We keep the constant tension and do not attribute it to walls. (ii) *A pre-bounce contraction.* Bouncing cosmologies usually solve the horizon problem with a long contracting phase. We do not adopt one: the framework starts from a false vacuum that crystallizes, not from a previous universe, and a contracting pre-history would move the initial-condition question one cycle back while adding its own problems (growth of anisotropy and curvature, entropy carried across the bounce). We prefer to carry the horizon problem openly (Sec. 5.1).

Data and code availability. Code, figures, build instructions and download links for all datasets used are at https://github.com/AIMFIRST-VN/big001. ## **Epilogue: Three Questions, Three Answers**

We set out with three questions and a pool table. Here is where the balls came to rest.

How does it work? The universe is the Big Apple: concentric rings of one kind of ball, a Planck-mass relic, each ring under a different law, and we live in the core with the seeds — Kernel

0,0,0, the only ring with light. Picture the moment before the first moment: a false vacuum, tense as a drawn bow. A bubble of true vacuum nucleates, its wall catalysing more bubbles, until the whole patch is a jammed crush of Planck-mass spheres pressed to the random-close-packing limit — some 10^{88} of them in what will become each Hubble volume, if the grinding went all the way to the Planck scale. Nothing can fall further. A loop-quantum bounce fires and the pile rebounds — not as an explosion but as an unloading: a rarefaction wave runs in from the surface and releases the ball layer by layer, the skin first and fastest, then the flesh, while the core is held by pressure and pounded by the reflected waves. That is H8 (Precious Rings – Dante's Purgatory): one kind of ball, one number for each, and a different law on every ring, the ring order being the original depth. In the core the balls collide: two Planck relics that merge are heavier than a relic is allowed to be, and the merger boils away into light, over and over, until the mergers stop keeping pace with the expansion — a depletion with no way back — and what is left is a bath of light with a cold trace of surviving spheres in it. That bath is the hot Big Bang; those survivors, the seeds, are our dark matter. No inflaton, no reheating, no residual vacuum: the light came from the relics. The flesh is a dark mantle that remembers nothing. The skin — 88% of the mass in both toys — is the dark rings around us: sorted by speed, the only ring that still keeps the geometry and spin of the bang, and the one we will never see except as a fossil in the initial conditions. Inside the core the crush is *assumed* to have been everywhere and to have no centre we can reach: Geometry A, homogeneous, its edge beyond the horizon. Seen as a whole, the rings are one centred event: Geometry B, in which the fastest went furthest and that ordering *is* a ballistic Hubble-like law, $v = r/t$, plus the tension below — at the price of having, as yet, no bounce mechanism for a finite cloud that sits 10^{40} inside its own horizon. Either way the debris does not push the universe faster; nothing outside a volume ever can, and nothing inside it pushes unless it has negative pressure. The acceleration we measure comes from a constant tension of space, $w = -1$, adopted and not derived. So the model works as far as the matter goes, hands the expansion to a tension it does not explain, and carries its open problems in the open: the number of survivors is one relation, $f = T_{\mathrm{eq}}/T_{\mathrm{conv}}$, that the grinding must *meet* rather than a number it derives, and the geometric cross-section misses it by up to twenty-eight orders of magnitude; the baryons must be imported; the thermal ring around the Cold Spot is not there; and the bounce is only meaningful without qualification if the jammed patch was the whole universe rather than a ball inside its own horizon.

Where are we? In Kernel 0,0,0, says H8 (Precious Rings – Dante's Purgatory) — the core of the Big Apple, the only ring with light, at or near its centre — and this is the question that tests it. Inside a homogeneous core a spherical flesh and skin exert nothing, so the placement predicts precisely what ΛCDM predicts, and the test can only falsify it, never single it out. If instead we sat off-centre in a centred cloud, the signature would be P4 (Off-Centre Flow): an off-centre dipole $\boldsymbol{\xi}$ in the structure-subtracted galaxy velocity field above the mock noise floor — the 95th percentile of the false centres that isotropic mocks manufacture on the same sky mask, about 2.5 Mpc at present — or a bulk flow that grows rather than fades with depth. We looked, and after the mock-mask and 2M++ controls the answer so far is: no offset claimed, two megaparsecs at most, with a 1.7 Mpc residual inside the mock distribution that one re-run will confirm or dissolve as calibration. What we do have is a bulk flow of 428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc, which the kinetic Sunyaev–Zel'dovich data would forbid only if $\gtrsim 250$ km s^{-1} of it persisted to a gigaparsec, and a sky whose extreme spots all lie in the southern hemisphere. There is a second handle, and it has a sign. Grinding is more complete where the crush was densest, so the core should be hotter and less dark at its centre than at its edge: primordial temperature falling outward, dark-to-baryon ratio rising outward — P12 (Kernel Gradient). The data say the core is flat in both to a few percent across everything we can see, so the gradient is saturated inside our horizon; a trend of the *opposite* sign would put us somewhere the rings say we cannot be. The reading we adopt is that we are not visibly off-centre, exactly as a position at the centre of Kernel 0,0,0 requires — and exactly as ΛCDM requires — and that the lopsidedness of the sky is more plausibly the work of a neighbour than of our own position. Where we are remains the least answered of the three questions, and the one whose answer would settle the most.

Where is the second Big Bang? Beyond the skin. Follow the coldest patch of the oldest light: galactic longitude 203ř, latitude −56ř, the Cold Spot. If it is the fingerprint of a neighbouring Big Pool Bang written into the initial conditions of our last-scattering surface, that neighbour sits about one billion light-years past the edge of everything we can see, with a mass of order 10^{48} kg (a factor of four up, two and a half down) — a dwarf beside the 2×10^{52} kg of survivors in each of our own Hubble volumes. It lies beyond our horizon, so nothing it has done since can reach us; its induced drift of our neighbourhood is a tenth of a kilometre per second. Seven other extreme spots lie on the same side of the sky, a candidate pool of neighbours clustered rather than surrounding us — though whether a neighbour can lopside the sky's power is not something we have computed. The neighbour reading makes promises it can be held to: no curl-mode fingerprint at the spot, where a vortex would leave one — P8 (Curl Test), which decides between the two readings; a few microkelvin leaked across the whole sky in a pattern fixed by the spots — P9 (Low-ℓ Tail), judged against a thousand Gaussian skies; and nothing contributed to the bulk flow — P10 (Induced-Velocity Bound), which it does not. The neighbour hypothesis will not survive as a story; it will survive, or fall, as the set of numbers in Table 7.

What surprised us

Seven things we did not expect when we racked the balls, and two things we were glad to lose.

A one-sided sky. All eight of the most extreme spots in the Planck map lie in the southern galactic hemisphere, none in the north, and the Cold Spot is one of them.

A spiral in the Cold Spot. The Cold Spot has the most coherent spiral winding of any of the 384 sky positions we scanned, a slow 40ř twist from centre to rim: what a frozen vortex would look like, and what a Gaussian fluctuation has no reason to do.

Neighbours push, but not outward on average. Surrounding bangs pull on our galaxies, harder toward them and inward in the gaps between them, but Gauss's law makes the sky-averaged pull exactly zero at every radius, and the directional part is about 10^{-7} of the Hubble rate. Only mass inside our own volume can push isotropically, and the kinetic Sunyaev–Zel'dovich gate caps what our own ejecta can contribute at about 1% of the observed acceleration. That is why the paper had to hand the acceleration to a constant tension of space.

Equal-mass neighbours must be very far away. If every bang had our own mass, a neighbour at the Cold Spot distance would burn a 4 K hole in the microwave sky instead of a 168 μK dimple. So either the Cold Spot source is a bang some 10^5 times lighter than ours or it is something else, and equal-mass bangs must keep a spacing of at least a thousand billion light-years, so that any global spin they induce is capped near 10^{-8} of the Hubble rate.

The neighbour is beyond the horizon. If it exists, nothing it has done since the bounce has reached us. Everything we can test is a fossil potential laid down before the first light.

The rings solve the budget. We spent a long time trying to make one history do two jobs — a jammed rebound whose ejecta are the dark matter, and a radiation era that leaves cold relics — and the energy budget refused every version of it. The answer was not a new ingredient but a map: the same relics, different physics by ring. We live in Kernel 0,0,0, the only ring with light, which is why there is anyone here to notice.

The ball peels. When we replaced the billiard balls with a fluid — a hard-sphere gas with a jamming pressure, in one dimension — the rebound did not explode. It unloaded from the outside in: the atmosphere left first, then the middle, and the core stayed and was recompressed by reflected waves five to nine times. The fluid sorted itself into three bands with edges at 45% and 98% of the mass at every kick strength — the three tissues of the apple — and needed a kick of *twice* the escape speed to unbind the 88% that the collisionless balls unbind at 0.7 times it. The difference is the pressure, which is what the rings switch on and off.

What we got rid of: the dark-matter particle, and the inflaton

The largest simplification the framework buys is the one it was built for. Dark matter here is not a new particle. It is the Planck-mass relics that survived the grinding in Kernel 0,0,0 — the cold trace left when the mergers that made the light stopped keeping pace with the expansion (H8 (Precious Rings – Dante's Purgatory)) — ordinary gravitating mass with no charge, no light, and no reason to appear in any detector but a gravitational one; the 88% that the rebound threw outward is the same stuff, but it is the dark rings outside Kernel 0,0,0, not the dark matter in our galaxies. Forty years of direct searches have found nothing, and in this picture that is the expected result: P5 (No Particle DM) turns the null into a prediction, and a confirmed particle detection would falsify H1 (Bounce – The Big Pool Bang) outright. What the framework does *not* get rid of is dark energy: the constant tension of space is adopted, its value is not derived, and Section 6 says so. The second thing we lost is comic inflation. There is no inflaton, no potential to tune, no slow roll and no eternal-inflation multiverse of bubbles: the universe does not inflate, it bounces, and the bounce is what lets it start without a singularity and without the Borde–Guth–Vilenkin past boundary. The price is stated as plainly as the gain: inflation was invented to make the sky flat and uniform, nothing in this framework yet does those two jobs, and the red tilt is a fitted number, not a derived one. So the scorecard reads: dark-matter particle gone, inflaton gone, dark energy kept, two of inflation's chores still waiting for a mechanism, and the jamming bounce and cold dark matter reconciled by rings — with the conversion that the reconciliation rests on still to be computed, and a cross-section gap of up to twenty-eight orders of magnitude standing in its way. That is the honest scorecard.

When we will know more

Most of the decisive tests are archival and cheap. The curl search (P8 (Curl Test)) at calibrated sensitivity and the low-multipole template (P9 (Low-ℓ Tail)) can be run on Planck PR4 products within a year; a null on the first retires the vortex reading, a null on the second the population reading. The JADES handedness control field is a single reprocessing of public JWST data. The kernel gradient (P12 (Kernel Gradient)) is already bounded by the SZ temperature–redshift relation and the cluster baryon fractions, and every new high-redshift cluster sample tightens it. Our own position (P4 (Off-Centre Flow)) waits on the next CosmicFlows release and on DESI and Euclid peculiar velocities, roughly 2027–2029, which should settle whether the 1.7 Mpc residual is calibration — and whether we sit at the centre of Kernel 0,0,0 as H8 (Precious Rings – Dante's Purgatory) says. The cold-relic statement, P11 (Cold Relics), passes every warm-dark-matter limit and constrains warm dark matter, not ΛCDM. The polarization ring and cyclonic B-modes, P1 (Polarization Ring) and P2 (Cyclonic B-modes), need degree-scale polarization at the microkelvin-arcminute level: Simons Observatory results toward 2028, CMB-S4 and LiteBIRD in the early 2030s. The gravitational-wave background from a first-order transition, P3 (Transition GW Background), is a LISA-era question, after 2035. By the end of this decade, then, the Cold Spot will be a vortex, a neighbour, or neither, and we will know whether we sit visibly off-centre in our own Big Pool Bang.

In one paragraph: a jammed crystal that bounced and peeled, a speed-sorted debris field arranged in rings — the core, Kernel 0,0,0, ground into the light we live in and the cold trace we call dark matter; the flesh and skin dark, and the seed of the Hubble flow or the Hubble flow itself depending on which ring one asks — and a cold fingerprint on the far wall that may belong to someone else. Each of these is stated so that a measurement can remove it.

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Appendix A. Exploratory Null Results and a Pre-registered Test

This appendix records the exploratory battery run on the eight most extreme large-scale spots of the Planck SMICA map ($N_{\rm side} = 128$, $|b| > 25$ř, 2ř smoothing; per-spot amplitude ΔT and half-maximum radius θ; positions in Sec. 11). Nulls are 60–300 Gaussian realizations drawn from the map’s own pseudo-C_ℓ and passed through the identical pipeline unless stated otherwise.

Locus test. If neighbours shared a common distance shell and formation density, their spots would follow a locus $|\Delta T| \propto \theta^3$, whereas Gaussian peaks (Bardeen et al. 1986) have amplitude and size nearly independent. A log–log fit gives slope -0.00, $r = -0.01$, 27th percentile of the Gaussian null ($+0.07 \pm 0.10$): no correlation, but the test is powerless as implemented (radii quantized to 2.5ř–5.5ř, amplitude spread $\sim$10%) and is reported as unresolved, not as an exclusion.

Internal dipoles and spin budget. Monopole-plus-dipole fits within 6ř in the local tangent frame: the eight extremes split 4/4 hot/cold with rank-matched amplitudes summing to $-11\,\mu$K against a mean $|\Delta T| \approx 162\,\mu$K (7% of Gaussian skies as balanced); opposite-sign spots lie 4.0ř closer than same-sign (null $+5.2$ř $\pm\,16.3$ř, $p = 0.30$); nearest-neighbour spacing is no more regular than Gaussian. Converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{n} \times \mathbf{d}$), the Cold Spot's internal dipole ($16.4\,\mu$K/deg) is the largest by a factor 2.3 and the others weakly co-align. These fits are dominated by $\ell \sim 2$–20 gradients that Gaussian peaks generically carry, so they are upper bounds on rotation, not spin measurements.

Spiral winding. The azimuthal phase of the $m = 1$ mode in 1.5ř rings from 1.5ř to 15ř around the Cold Spot is locked ring-to-ring with coherence 0.988 — above all 300 random positions (null 0.72 ± 0.18) — and drifts by $\sim$ 3ř of phase per degree of radius ($\sim$ 40ř total twist). A static background gradient gives high coherence but zero winding; only the winding is spin-specific. The $m = 2, 3$ modes are unremarkable, and this is a post-hoc test.

Curl search. The public PR4 lensing release (Carron, Mirmelstein & Lewis 2022) contains only gradient modes, so we implemented an approximate temperature-only quadratic estimator ($N_{\rm side} = 256$, $\ell \leq 512$, spin-1 curl projection, mean field from 40 Gaussian simulations; unnormalized). Stacking curl-potential variance in 6ř discs, the eight spots sit at the 82nd percentile of 60 simulated skies and the Cold Spot at the **38th** — indistinguishable from a random position. This does not trigger the retirement clause of P8 (Curl Test), which requires a sensitivity reaching the temperature-dipole-implied rotation that this reduced-resolution estimator does not attain.

Polarization. On the SMICA Q/U maps ($N_{\rm side} = 256$, $\ell \leq 512$, E/B decomposition, 1ř smoothing): stacked B-mode variance in 6ř discs at the eight spots is at the 66th percentile of random positions, E-mode variance at the 35th, and the Cold Spot's radial-E profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK — the predicted 1–3 μK ring of P1 (Polarization Ring) is neither detected nor excluded.

Lensing convergence. The sign-weighted PR4 κ stack (Planck Collaboration 2020 VIII; Carron et al. 2022) on the eight positions is $+0.006$ against a random-position null of 0.000 ± 0.013 (4/8 sign matches): no integrated mass scars. The Cold Spot's inner-2ř $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) has the sign of its known shallow foreground supervoid; the 8ř–20ř annulus never exceeds $+0.6\sigma$, so there is no mass "belt". A stack on $\sim$27,000 ordinary small spots shows cold spots on κ deficits (-7.2σ) and hot on excesses ($+2.2\sigma$), but quadratic estimators leak temperature extrema into the mass map; a polarization-only κ stack is the required discriminating check.

CMB-independent tracers. Against the Quaia Gaia–unWISE quasar catalogue (Storey-Fisher et al. 2024; 756k quasars, $z \sim 1$–3, selection-function corrected) the small-spot correlation vanishes (cold -1.1σ, hot -0.3σ) and the eight extremes are within $\sim 1\sigma$ of random (Cold Spot -0.018, 0.6σ); this is not a sensitive null, since Quaia's kernel barely overlaps the ISW kernel. Against WISE$\times$SuperCOSMOS (Bilicki et al. 2016; 17.7M galaxies, $z = 0.1$–0.4, high-passed at 10ř): small spots remain null (cold -0.7σ, hot $+0.2\sigma$), while the extremes show seven of eight sign matches and a sign-weighted stack of $\approx +2.7\sigma$ against a correlated-noise null from 300 random 5ř discs. Qualifications: the result rests on three sightlines; the nominal binomial 3.5% ignores the multi-tracer, multi-statistic search (global $p \sim$ 10–30%); extremeness-conditioning inflates the correlation; the 10ř high-pass self-subtracts genuine supervoids; and extinction/stellar residuals at $|b| = 25$–40ř are of order the per-spot signals. Verdict: a 1–2σ-class hint, to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30$ř and extinction regression. The 2M++ cross-match of the eight sightlines ($< 180\,h^{-1}$ Mpc) matches spot signs at coin-flip rates (5 of 8).

JWST handedness. The JADES field at (224ř, −54ř), 8.6ř from the Cold Spot, shows a claimed 2 : 1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025); the sample is small, the method contested, and a Doppler boost from the Milky Way's rotation is a mundane candidate. The discriminating measurement is a control field ($>$ 100ř away): a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. The local anchor is already measured: the same-handedness angular correlation over 77{,}840 SDSS galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin, with a 1.9σ global excess.

Coherence-gradient hypothesis (adopted by H8 (Precious Rings – Dante's Purgatory)). In the N-body toy, only the earliest, fastest ejecta escape in a single pass and preserve geometric correlations; everything slower undergoes violent relaxation, which conserves spin magnitude but randomizes spin geometry. If any residual radial ordering survived in our Hubble volume — H5 (Our Position – Where Is Nemo) — one would expect a *coherence gradient*: fossil geometry intact in the earliest-ejected material, degrading toward the later ejecta. Intermediate tracers (quasar-polarization alignments; galaxy handedness axes) lie 24ř–73ř from the Cold Spot direction, consistent with random axes, and the local same-handedness correlation is zero to a few parts in 10^3. H8 (Precious Rings – Dante's Purgatory; Sec. 3) makes this its statement — free streaming preserves initial geometry and spin in the shell, collisions in Kernel 0,0,0 and orbit mixing in the mantle erase them — with the consequence that the coherent material is outside our horizon and the material we can survey is the mixed material; the local nulls are what it predicts. It has no validated observable of its own inside Kernel 0,0,0.

Statistical accounting. Across the battery (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, winding coherence, κ stack, curl percentile, tracer stacks) the individual results are $p \sim 0.06$–0.7; under any global look-elsewhere correction the sub-2σ items are jointly consistent with noise. The Gaussian nulls use the unmasked pseudo-C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals, and the spots were selected on the same map on which all statistics were evaluated, so the hypotheses were partly formulated post hoc. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects.

Pre-registered commitment (P8 (Curl Test)). The load-bearing, falsifiable claim of the vortex reading is the curl test. We commit in advance that a clean stacked curl-mode null at the Cold Spot, at a sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation in favor of the neighbour reading (H3 (Neighbour – Big Pool Bang Multiverse 2)/H6 (Neighbour Location)); a localized curl detection there makes the Cold Spot a relic vortex of our own bang.

Appendix B. A Fixed Grid of Surrounding Bangs (Consistency Check)

The population model in its most symmetric form — the limiting case of the one-sided population H7 (Pool Model) adopts (Sec. 11e) — places Big Pool Bangs on a fixed lattice with our own at one site. As a consistency check, we ask whether the neighbours' pull on our ejecta could contribute to the observed late-time acceleration. Point masses of equal mass were placed on a cubic grid of unit spacing (26 neighbours for $3 \times 3 \times 3$, 124 for $5 \times 5 \times 5$); test galaxies were placed at distance r from us in random directions and the vector sum of the neighbours' inverse-square pulls was evaluated (Table 8). Units: the pull of one neighbour on us.

Table 8. Net pull of a cubic lattice of equal-mass neighbours on test galaxies at distance r from us.

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.1	0.0000	+0.012	−0.008
0.2	0.000	+0.10	−0.06
0.3	0.000	+0.34	−0.20
0.4	0.00	+0.85	−0.44
0.45	0.00	+1.26	−0.60

The results (Table 8) are identical for both grid sizes. Galaxies lying toward a neighbour are pulled outward with a pull that grows steeply with r; galaxies lying toward the gaps between neighbours are pulled inward; the sky average is zero at every radius. This is Gauss's law, not a property of the arrangement: the sky-averaged radial acceleration at radius r is $GM_{\rm enc}(r)/r^2$, and external sources enclose nothing. Only mass, or negative pressure, *inside* the observed volume can change the isotropic expansion rate. The lattice therefore predicts (i) zero contribution to the monopole acceleration, and (ii) a direction-dependent expansion rate with the cubic-harmonic ($\ell = 4$) pattern of the lattice, fast toward the six nearest neighbours and slow toward the eight diagonal gaps. Observation (ii) is a further falsifier for the lattice limit of H7 (Pool Model; Table 7): supernova and BAO expansion-rate anisotropy is limited to the few-percent level with no cubic pattern, while the amplitude implied by the neighbour mass and distance of Sec. 10 is $\sim 10^{-7}$ at the $r \approx 0.3$ reached by the supernova sample. The one form in which surrounding mass raises the local expansion rate isotropically is a local underdensity, where the effect comes from the mass *missing inside.* A $\delta \approx -0.3$ underdensity within $\sim$300 Mpc (Keenan, Barger & Cowie 2013; disputed by Wu & Huterer 2017 and Kenworthy, Scolnic & Riess 2019) would imply an outflow of $\sim$1000 km s^{-1} at its edge, in tension with the kSZ gate of Sec. 6; this paper takes no position on whether the void exists, only that a Gpc-scale void deep enough to mimic Λ is excluded by the kSZ gate. This appendix is a consistency check on the adopted expansion history of Sec. 6, not its basis.

Appendix C. Kinematic Mimicry of Acceleration: Tests

This appendix records the tests summarized in Sec. 6 of whether the ejecta distribution could mimic acceleration for an interior observer, and the toy $w(z)$ drift.

Relativistic (LTB) recalculation. Dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells; what the Newtonian shell toy below ("The $w(z)$ drift") gets wrong is the *observables*. Integrating radial null geodesics through the evolving LTB metric (300 shells seeded from the ejecta speed distribution, fallback mass virialized into the interior, $d_L = (1+z)^2 R$) gives $q_0^{\rm eff} = +0.16$ ($w_{\rm eff}(0) \approx -0.23$): mild deceleration, no apparent acceleration, with a distance-modulus shape ~ 0.35 mag from both ΛCDM and coasting out to $z \sim 5$. The lightcone threads only the free-streaming 9%, and shell crossing was not monitored.

Profile inversion. Since the ejection profile is an input, we searched power-law, bimodal fast-tail, and stochastically lumpy profiles for any that accelerates. Smooth families robustly decelerate ($q_0^{\rm eff} \sim +1$ to $+4$); apparently negative values in lumpy configurations are seed-ensemble noise (± 1.4, exceeding the target $|q_0| \approx 0.55$, so that family is unconstrained rather than excluded). The known way an LTB model mimics acceleration — the giant void, observer inside a Gpc-scale underdensity — is the *opposite* density ordering from outward ejection.

Blast wave and the kSZ gate. A rebound acting as a blast wave would sweep ejecta into a shell and leave a rarefied cavity, the giant-void geometry; swept-shell profiles do give smoother lightcones

(0.03–0.08 mag residuals vs ~ 1 mag). This is not a rescue: a 0.08 mag residual corresponds to $\Delta\chi^2 \sim 400$ against Pantheon+, a cavity deep enough for $q_0 \approx -0.55$ is the giant-void model excluded by kinetic Sunyaev–Zel'dovich and Compton-y constraints (Zhang & Stebbins 2011), and mass swept into a distant shell is unavailable as local dark matter. Imposing the constraints first: the kSZ limit on coherent cluster flows ($v \lesssim 300$ km s^{-1} at Gpc distances) caps the mean interior contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) to $\lesssim 0.01$ (3 Gpc), i.e. $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy; this is the result quoted in Sec. 6.

The $w(z)$ drift. Integrating the measured ejecta speed distribution as cold spherical shells decelerating in their enclosed mass, and computing the equation of state an interior observer would fit, gives w_{eff} drifting from -0.29 at $z = 2$ to -0.31 today (Fig. 5): a few-percent drift near the coasting value $-1/3$. Because this apparent equation of state is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy, it cannot be constant. The DES-SN5YR + DESI DR2 BAO (+CMB) w_0w_aCDM fit prefers evolving w at $\sim 3\sigma$ ($w_0 \approx -0.8$, $w_a \approx -0.7$; DES Collaboration 2024; DESI Collaboration 2025), with w becoming *less* negative with time; the toy shells drift the *opposite* way. The remap is crude (91% of the ejecta re-bind in a static enclosed-mass model, so the flow is built from the fastest 9%), but as it stands the kinetic sector shares with the data only the qualitative statement that w evolves, and predicts the wrong sign of the drift.

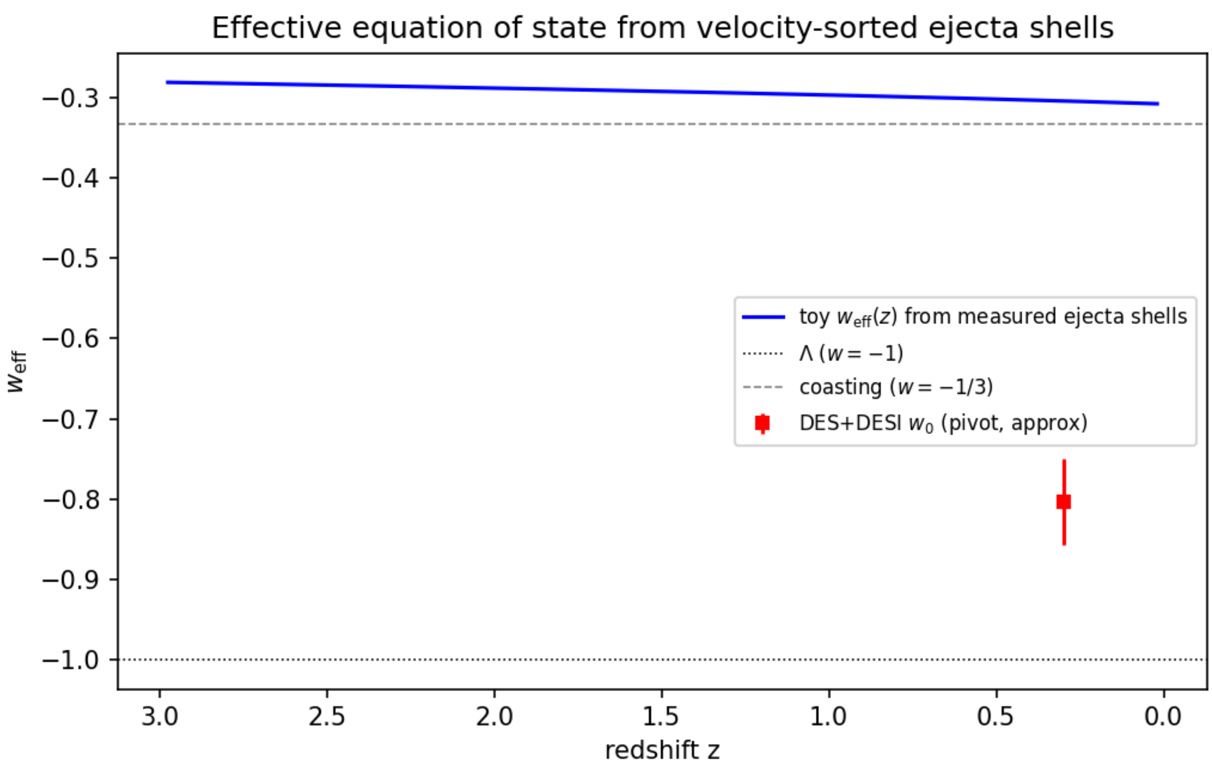

Figure 5: Fig. 5 — Toy effective equation of state from velocity-sorted ejecta shells: a few-percent drift near $w = -1/3$, in the opposite direction to the DES+DESI w_0w_a best fit, and of insufficient magnitude.

A framework-native candidate, with its cost. If crystallization is incomplete today, a residual unconverted vacuum fraction would be smooth, non-clumping, and immune to the kSZ gate —

qualitatively dark energy. Quantitatively this re-imports the cosmological-constant problem verbatim (tuning to one part in 10^{123}), and ongoing conversion would imply present-epoch nucleation, domain walls (Zel'dovich, Kobzarev & Okun 1974), and an unconverted component at BBN and recombination — none computed here; it is rejected. The boiling picture does anchor H3 (Neighbour – Big Pool Bang Multiverse 2)–H4 (Population – The Dark Comic Background) in Coleman & De Luccia (1980) bubble nucleation, whose collisions imprint disc-shaped CMB features with specific polarization profiles (Aguirre & Johnson 2011) and, generically, source the stochastic gravitational-wave background of P3 (Transition GW Background).

Appendix D. N-body Ejection Details

This appendix records the toy N-body results summarized in Sec. 4 (`nbody_ejection.py`, `nbody_turbulent.py`, `ejecta_analysis.py`). The runs illustrate the ejection *mechanism*; the ejected fraction is the mass of the dark rings under H8 (Precious Rings – Dante's Purgatory), not the local dark-matter fraction.

Ejection fraction (toy N-body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial kick $v = f\, v_{\rm esc}$ (with $v_{\rm esc}$ the surface escape speed) has a steep ejection threshold: $f = 0.6 \to 3\%$, $0.8 \to 41\%$, $0.9 \to 98\%$, $f \geq 1 \to 100\%$. A monolithic kick reaches an 88% ejected fraction only in the narrow window $f \approx 0.85$–0.9, and the surface-escape-speed kick ($f = 1$) ejects everything. A Maxwellian turbulent component of dispersion $\sigma_{\rm rel}\bar{f}v_{\rm esc}$ superimposed on a mean rebound $\bar{f}v_{\rm esc}$ broadens the transition: at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\rm rel} = 0.5$) through $\approx$ 88% (1.0) to 98% (1.5). Over a 15-seed ensemble at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, at a $\sigma_{\rm rel}$ the cascade model of Sec. 2 only marginally reaches (0.65–0.78); $(\bar{f}, \sigma_{\rm rel})$ remain inputs to be derived from the bounce turbulence spectrum. The fluid toy of Sec. 3.2 reaches the same 88% at $\bar{f} = 2.0$ without turbulence.

Ejecta kinematic structure (15 seeds, 13{,}140 ejecta). Ejection is extended (median ejection time $\sim$1 dynamical time; 10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71\, v_{\rm esc}$ for the earliest ejecta to $0.28\, v_{\rm esc}$ for the latest (Fig. 2), so radial position maps onto ejection epoch. The tangential structure shows no correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) matches shot noise (0.053 ± 0.024), and early-vs-late shell axes align only weakly ($\langle \cos\theta \rangle = 0.37 \pm 0.49$). The isotropic cascade therefore supplies no preferred axis; if one exists it must be seeded externally (Sec. 9).